

Digital Signal Analysis and Processing

Complete Exam Notes

CT 704 (BEI/BCT, IV/I) · EX 753 (BEX, IV/II)

Institute of Engineering, Tribhuvan University

The **whole CT 704 syllabus**, all seven chapters and all 45 lecture hours, built from the syllabus, five lecturers' note sets and **all 45 past papers (2066–2082 BS)** across both course codes. Every topic carries a frequency tier counted from those papers, and every count comes from `tools/ch_tally.py` reading the Sorted PYQ document, not from memory.

Worked problems live in the companion volume, **Digital Signal Analysis and Processing – Complete Numerical Problems & Solutions**: **48 worked families** over the same seven chapters, each opening with the question exactly as its paper prints it.

Chapter 8 of the PYQ document is not here. That chapter is the EX 753 extras — DSP fundamentals, A/D quantization error, bit-serial and distributed arithmetic, pipelining, DSP processor chips — which lie *outside* the CT 704 syllabus and are asked only in the 17 EX 753 papers. It is **3.4 of 80 marks** overall. These notes are the other **78 of 80**.

Ch	Topic	Hours	Papers	Questions	Marks	Worked
1	Discrete-Time Signals and Systems	8	44/45	104	12.8	12
2	Z-Transform	4	40/45	51	7.4	7
3	Analysis of LTI Systems in the Frequency Domain	6	38/45	55	8.4	5
4	Discrete Filter Structures	8	43/45	68	10.0	6
5	FIR Filter Design	6	42/45	70	12.0	4
6	IIR Filter Design	6	44/45	50	13.1	7
7	Discrete Fourier Transform and FFT	7	44/45	99	14.3	7
Total		45		497	78.0	48 families

The marks column is normalised to one 80-mark paper, so it reads directly as “what this chapter is worth on the day”. Six procedures carry the subject: the convolution sum (45 questions), the DIT/DIF FFT (41 papers), the lattice step-down (38), the inverse Z-transform (32), the bilinear Butterworth design (32) and the circular convolution (35). Chapters 7, 6, 1 and 5 together are **52 of the 80 marks**, and **Chapter 2 is the best marks-per-hour in the subject** — 7.4 marks for 4 hours, from two questions that are always asked together.

Contents

1	Discrete-Time Signals and Systems	3
2	Z-Transform	15
3	Analysis of LTI Systems in the Frequency Domain	21
4	Discrete Filter Structures	27
5	FIR Filter Design	33
6	IIR Filter Design	38
7	Discrete Fourier Transform and FFT	45
	Abbreviations	51

1. Discrete-Time Signals and Systems

8 hrs · the largest chapter, **105 questions across the 45 papers** · worth about **12 of 80 marks** every year · 10 syllabus sub-topics · numericals in the Ch1 companion.

Read this first:

- **Convolution is the single most-asked question in the whole subject:** some form of “find $y(n)$ given $h(n)$ and $x(n)$ ” appears in **34 of 45 papers** **TOP 34/45**, usually for 5 or 6 marks. If you learn one thing in this chapter, learn that.
- The chapter’s other four regulars, in order:
 - energy Vs power signal + is it periodic **TOP 10/45**,
 - test a given system for linearity / time-invariance / causality / stability **HOT 7/45**,
 - find the fundamental period of a given sequence **HOT 7/45**,
 - sampling an analog signal and writing down $x[n]$ **HOT 7/45**.
- Counted by *topic* rather than by exact question, system properties come up in **16** papers and sampling in **10**, so nothing here is optional.
- **The marks sit in the method, not the answer.** Almost every question in this chapter is a 3-line recipe applied to new numbers. Sections 1.3, 1.4, 1.8 and 1.9 each end with the recipe written out as steps: memorise those, not the worked examples.
- Tier chips count papers out of 45: **TOP n/45** ≥ 10 papers, **HOT n/45** 4–9, **PIN n/45** 1–3. **Bold** year codes are Regular exams, plain are Back; typewriter codes are EX 753 (BEX) papers.

1.1 What a discrete-time signal is, and how it is classified

Define and compare continuous-time and discrete-time signals; classify a given signal **2 3**

PIN 3/45 69 Bh, 73 Ch, 76 Ch · never asked alone, always as the 2-mark opening of a bigger question

By the independent variable

- **Continuous-time (analog):** $x(t)$ defined for a continuum of t .
 - Eg speech voltage at a microphone, video signal.
- **Discrete-time:** $x[n]$ defined only at discrete instants, n integer.
 - Instants need not be equally spaced, but in practice $t = nT$ is used.
 - Eg monthly sales of a corporation, daily stock-market average.
 - $x[3.12]$ **does not exist**. A DT signal is a *sequence*, not a function sampled finely.

By the value the signal takes

- **Continuous-valued:** amplitude takes any value in a range.
- **Discrete-valued:** amplitude comes from a finite set.
- **Digital signal** = discrete in *time* + discrete in *value*.
 - This is the only kind a processor can store, so A/D does both steps: sampling (Ch 1.10) then quantization (Ch 8).
- **Exam trap:** “discrete-time” and “digital” are not synonyms. Sampling alone gives a discrete-time, continuous-valued signal.

Define even and odd discrete-time signals; find the even and odd part of a given signal 2 3 4+5

HOT 4/45 71 Ch, 70 Asa, 71 Shr, 67 Mng · plus 76 Ch which asks the definition then a plot · worked instances in Ch1-num §7

Definitions

- **Even (symmetric):** $x[-n] = x[n]$.
- **Odd (antisymmetric):** $x[-n] = -x[n]$.
- An odd signal must have $x[0] = 0$, since $x[0] = -x[0]$.
- Any signal splits into exactly one even + one odd part.

The two formulas

$$x_e[n] = \frac{1}{2}(x[n] + x[-n]), \quad x_o[n] = \frac{1}{2}(x[n] - x[-n])$$

Derivation, if the question says “show that”

- Start from $x[n] = x_e[n] + x_o[n]$ (1)
- Put $n \rightarrow -n$: $x[-n] = x_e[-n] + x_o[-n]$ (2)
- Use $x_e[-n] = x_e[n]$ and $x_o[-n] = -x_o[n]$ in (2): (3)
 $x[-n] = x_e[n] - x_o[n]$
- (1)+(3) $\rightarrow x[n] + x[-n] = 2x_e[n]$.
- (1)-(3) $\rightarrow x[n] - x[-n] = 2x_o[n]$.

Method, every time

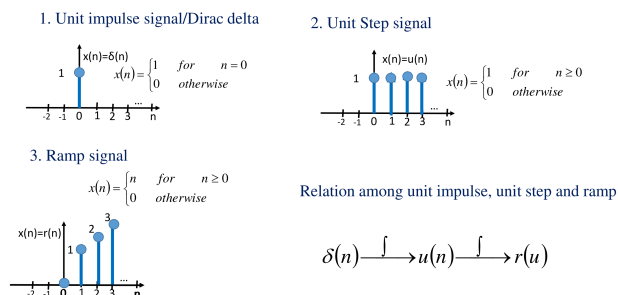
1. Write $x[n]$ as a sequence with the $n = 0$ sample marked.
2. Fold it to get $x[-n]$: same samples, index sign flipped.
3. Half the sum, half the difference, aligning on n .
4. **Check:** x_e must be symmetric about $n = 0$ and $x_o[0] = 0$.

1.2 Elementary discrete-time signals

Define and plot the unit step / unit impulse; explain their relation 1+2 3

PIN 3/45 73 Ch, 74 Ch, 69 Bh · the plotting half is done in Ch1-num §8

Elementary discrete-time signals



The three building blocks and the chain that links them. BB Sir, deck 1.7

The five standard sequences

- **Unit impulse** $\delta[n] = 1$ at $n = 0$, else 0.
- **Unit step** $u[n] = 1$ for $n \geq 0$, else 0.
- **Unit ramp** $r[n] = n$ for $n \geq 0$, else 0.
- **Real exponential** $x[n] = \alpha^n$: decays for $|\alpha| < 1$, grows for $|\alpha| > 1$, alternates sign when $\alpha < 0$.
- **Complex exponential** $x[n] = r^n e^{j\theta n} = r^n (\cos \theta n + j \sin \theta n)$.

Relations worth quoting (this is the marks)

$$\delta[n] = u[n] - u[n-1], \quad u[n] = \sum_{k=-\infty}^n \delta[k]$$

- Step is the running sum of the impulse; impulse is the first difference of the step.
- Ramp is the running sum of the step: $\delta \rightarrow u \rightarrow r$.
- **Sifting:** $x[n]\delta[n-k] = x[k]\delta[n-k]$, so $x[n] = \sum_k x[k]\delta[n-k]$. **This one line is what makes convolution work (§1.9).**

1.3 Energy and power signals

Define / compare energy and power signals, and classify the given signal 2+2 3 5

TOP 10/45 79 Bh, 69 Ch, 81 Bh, 74 Ash, 79 Ba, 82 Bh, 72 Ka, 68 Bh, 75 Ch, 74 Ma · in 81 Bh only the periodicity half is asked · always paired with §1.4 · worked in Ch1-num §5

Over a finite interval $n_1 \leq n \leq n_2$

$$E = \sum_{n=n_1}^{n_2} |x[n]|^2, \quad P_{av} = \frac{1}{n_2 - n_1 + 1} \sum_{n=n_1}^{n_2} |x[n]|^2$$

Over all time

$$E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2, \quad P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

- **Energy signal:** $0 < E < \infty$, and then $P = 0$.
- **Power signal:** $0 < P < \infty$, and then $E = \infty$.
- **A signal can be neither** (Eg $x[n] = n$), and **never both**.

How to decide in one line

1. Finite-length or decaying sequence $\rightarrow$ energy signal.
2. Never-ending, bounded, non-decaying $\rightarrow$ power signal.
3. Every periodic signal is a power signal, with $P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$ over one period.
4. Every complex exponential $e^{j(\omega_0 n + \phi)}$ has $|x[n]| = 1$, so $P = 1$ and $E = \infty$: **power signal, no calculation needed**. That is the whole answer to 79 Ba and 82 Bh.

Standard results to quote

- $x[n] = \delta[n]$: $E = 1, P = 0 \rightarrow$ energy.
- $x[n] = u[n]$: $E = \infty, P = \frac{1}{2} \rightarrow$ power. **$P = \frac{1}{2}$, not 1:** the sum runs over $2N+1$ samples but only $N+1$ are non-zero.
- $x[n] = A$ (constant): $E = \infty, P = A^2 \rightarrow$ power.
- $x[n] = a^n u[n], |a| < 1$: $E = \frac{1}{1-a^2} \rightarrow$ energy.

Units

- Follow the signal: a voltage signal gives energy in V^2s and power in V^2 .

Sums you will need in every one of these (learn this block)

$$\sum_{n=-N}^N 1 = 2N+1, \quad \sum_{n=0}^N 1 = N+1$$

$$\sum_{n=0}^{N-1} a^n = \begin{cases} N, & a = 1 \\ \frac{1-a^N}{1-a}, & a \neq 1 \end{cases}$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \quad (|a| < 1), \quad \sum_{n=k}^{\infty} a^n = \frac{a^k}{1-a}$$

$$\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta}), \quad \sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$$

1.4 Periodicity of a discrete-time signal

Determine whether the given sequence is periodic; if it is, find the fundamental period 2+2 3 4 5

HOT 7/45 80 Bh, 80 Ba, 78 Bh, 70 Ch, 69 Bh, 67 Mng, 66 Ma · another 10 papers ask it as the second half of the energy/power question · worked in Ch1-num §6

Definition

- $x[n]$ is periodic with period $N > 0$ if $x[n] = x[n + N]$ for all n .
- Smallest such N is the **fundamental period**. N must be an **integer**: that is the whole difference from the continuous-time case.

The rule, and where it comes from

- For $x[n] = e^{j\omega_0 n}$ to repeat: $e^{j\omega_0(n+N)} = e^{j\omega_0 n} \rightarrow e^{j\omega_0 N} = 1$
 $\rightarrow \omega_0 N = 2\pi m$.
- So $\frac{\omega_0}{2\pi} = \frac{m}{N}$ must be a **rational number**.
- **A discrete-time sinusoid is periodic only if $\omega_0/2\pi$ is rational.**
- Then $N = \frac{2\pi m}{\omega_0}$ with m the smallest integer making N an integer.

Method, every time

1. Read off ω_0 (the coefficient of n inside cos, sin or the exponent).
2. Form $\omega_0/2\pi$. Irrational $\rightarrow$ **aperiodic**, stop.
3. Rational = m/N in lowest terms $\rightarrow$ fundamental period is N .
4. Sum of two periodic signals: $N = \text{LCM}(N_1, N_2)$, and it is periodic only if N_1/N_2 is rational.

The three traps the papers actually set

- $x[n] = \sin(0.8n)$ or $e^{j(7/5)n}$: ω_0 has **no π in it**, so $\omega_0/2\pi$ is irrational $\rightarrow$ **aperiodic (66 Ma, 69 Bh)**.
- $x[n] = \cos(\pi n^2/8)$: n^2 , not $n \rightarrow$ not a sinusoid, test $x[n + N] = x[n]$ directly. It is periodic, $N = 8$ (**70 Ch**).
- $\sin(\pi n/\sqrt{2} + \pi/8)$: the $\sqrt{2}$ makes $\omega_0/2\pi$ irrational $\rightarrow$ aperiodic (**67 Mng**).

Show that a discrete-time signal is periodic in frequency with period 2π 2 4

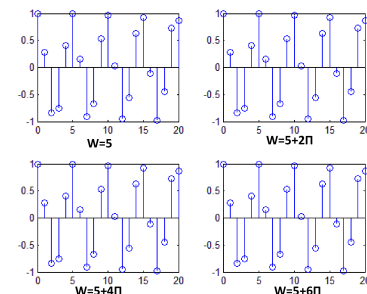
PIN 2/45 69 Bh, 80 Bh · also the reason every DTFT and DFT plot is drawn over one 2π span

- Raise the frequency by 2π :

$$e^{j(\omega_0+2\pi)n} = e^{j2\pi n} e^{j\omega_0 n} = e^{j\omega_0 n}$$

because $e^{j2\pi n} = 1$ for integer n .

- So $\omega_0, \omega_0 \pm 2\pi, \omega_0 \pm 4\pi \dots$ are the **same sequence**, sample for sample.
- Consequences worth one line each:
 - All distinct DT frequencies lie in one span of 2π , usually $-\pi \leq \omega < \pi$.
 - $\omega = \pi$ is the **highest** rate of oscillation (alternating $+1, -1$); $\omega = 0$ and $\omega = 2\pi$ are both DC.
 - In continuous time nothing like this happens: $e^{j\Omega t}$ is distinct for every Ω .



$\cos(\omega n)$ at $\omega = 5, 5 + 2\pi, 5 + 4\pi, 5 + 6\pi$: four different formulas, one sequence. BB Sir, deck 1.1

1.5 Transformation of the independent variable

Plot the transformed sequence, Eg $x[-2n + 3]$ or $u[n] - u[n - 3] + 5\delta[n - 4] + nu[n - 6]$ 2 3 3+2

HOT 4/45 76 Ch, 74 Ch, 66 Ma, 69 Bh · pure plotting marks · every instance is worked in Ch1-num §8

The three operations

- **Shift** $x[n - k]$: delay by k (move right) for $k > 0$; advance for $k < 0$.
- **Fold (time reversal)** $x[-n]$: mirror about $n = 0$.
- **Scale** $x[an]$:
 - $|a| > 1$ integer = **decimation**, keep every a -th sample, **samples are lost**.
 - $|a| < 1$ = interpolation, new samples must be filled in.
 - **Not reversible like the CT case**: this is why down-sampling needs its own treatment.
- **Amplitude operations**: addition, multiplication, scaling act sample by sample on aligned indices.

Order of operations for $x[an + b]$ (where marks are lost)

1. **Shift first, then scale/fold**, or you will shift by the wrong amount.
2. Safer still: work index by index. Set $m = an + b$, then for each sample $x[m]$ of the original, its new position is $n = (m - b)/a$.
3. Keep only integer n ; a non-integer means that sample disappears.
4. **Check one point**: put $n = 0$ into $an + b$ and confirm the value lands where you drew it.

Worked index map: $x[-2n + 3]$, $x[n] = \{1, 2, 0, -1, -3, -4\}$ starting at $n = 0$

- $n = 0 \rightarrow x[3] = -1$; $n = 1 \rightarrow x[1] = 2$;
 $n = -1 \rightarrow x[5] = -4$; $n = 2 \rightarrow x[-1] = 0$.
- Only odd-indexed original samples survive: folding + decimation by 2.

1.6 Discrete-time Fourier series (DTFS)

Explain how the Fourier series coefficients are calculated 3 4

PIN 2/45 73 Shr, 72 Ch · 76 Ash adds Dirichlet's conditions, see §1.7

The pair

- A periodic $x[n]$ of period N has fundamental frequency $\omega_0 = 2\pi/N$.

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} \quad (\text{synthesis})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n} \quad (\text{analysis})$$
- $\langle N \rangle$ means *any* N consecutive values, usually 0 to $N - 1$.
- $a_k = a_{k+N}$: only N distinct coefficients exist, because $e^{j(k+N)\omega_0 n} = e^{jk\omega_0 n}$.
- That is the one real difference from the continuous-time Fourier series, which has infinitely many distinct coefficients.

Deriving a_k , if the question says “explain the process”

1. Start from the synthesis equation.
2. Multiply both sides by $e^{-jr(2\pi/N)n}$ and sum over one period.
3. Interchange the two summations.
4. Use the orthogonality result

$$\sum_{n=\langle N \rangle} e^{j(k-r)(2\pi/N)n} = \begin{cases} N, & k - r = 0, \pm N, \pm 2N \dots \\ 0, & \text{otherwise} \end{cases}$$

5. Every term dies except $k = r$, leaving $\sum x[n] e^{-jr(2\pi/N)n} = a_r N$.

Method for a numerical (faster than the integral)

1. Find N first (§1.4), then $\omega_0 = 2\pi/N$.
2. Expand every cos/sin with Euler's formula.
3. Force each exponent into the form $jk(2\pi/N)n$ and read a_k off by inspection.
4. Use $a_k = a_{k+N}$ for anything outside $0 \leq k \leq N - 1$.

Properties of the DTFS ($x[n]$, $y[n]$ both period N)

Property	Signal	Coefficient
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time shift	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency shift	$e^{jM(2\pi/N)n}x[n]$	a_{k-M}
Time reversal	$x[-n]$	a_{-k}
Conjugation	$x^*[n]$	a_k^*
Periodic convolution	$\sum_{r=\langle N \rangle} x[r]y[n-r]$	$N a_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N \rangle} a_l b_{k-l}$
Parseval	$\frac{1}{N} \sum_{n=\langle N \rangle} x[n] ^2$	$\sum_{k=\langle N \rangle} a_k ^2$

Conjugate symmetry, asked as a rider: if $x[n]$ is real then $a_{-k} = a_k^*$. Real and even $\rightarrow a_k$ real and even; real and odd $\rightarrow a_k$ imaginary and odd with $a_0 = 0$. Parseval reads: total average power in one period = sum of the average powers $|a_k|^2$ of all harmonics.

1.7 Discrete-time Fourier transform (DTFT)

Explain the Fourier transform multiplication property for two sequences; write Dirichlet's conditions

4+3

PIN 1/45 76 Ash · the DTFT itself is assumed knowledge in every Ch 3 question, so this section is a prerequisite more than a target

The pair

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}, \quad x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

- Derived from the DTFS by letting $N \rightarrow \infty$: the discrete lines Na_k merge into the continuous envelope $X(e^{j\omega})$.
- $X(e^{j\omega})$ is periodic in ω with period 2π , because $e^{-jn(\omega+2\pi)} = e^{-j\omega n}$. So the inverse integral can be taken over any 2π span.
- Exists when $x[n]$ is absolutely summable, $\sum |x[n]| < \infty$.

Dirichlet's conditions (quote all three)

1. $x[n]$ is absolutely summable, $\sum_n |x[n]| < \infty$.
2. $x[n]$ has a finite number of maxima and minima in any finite interval.
3. $x[n]$ has a finite number of finite discontinuities in any finite interval.

Sufficient, not necessary: $u[n]$ fails condition 1 yet has a DTFT once impulses in ω are allowed.

Properties of the DTFT

Property	Signal	Transform
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(e^{j\omega}) + bX_2(e^{j\omega})$
Time shift	$x[n - n_0]$	$e^{-j\omega n_0} X(e^{j\omega})$
Frequency shift	$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega-\omega_0)})$
Time reversal	$x[-n]$	$X(e^{-j\omega})$
Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
Convolution	$x[n] * h[n]$	$X(e^{j\omega})H(e^{j\omega})$
Multiplication	$x_1[n]x_2[n]$	periodic convolution of the spectra, $\div 2\pi$
Parseval	$\sum_n x[n] ^2$	$\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$

If $x[n]$ is real, $X(e^{j\omega})$ is conjugate symmetric: magnitude even, phase odd. Real and even $\rightarrow$ transform real and even; real and odd $\rightarrow$ transform imaginary and odd. **Parseval here says energy, not power**: it is the DTFS version that talks about power.

Multiplication property, and why it is not a plain product

- Convolution in one domain = multiplication in the other, both ways:

$$x[n] * h[n] \longleftrightarrow X(e^{j\omega})H(e^{j\omega})$$

$$x_1[n]x_2[n] \longleftrightarrow \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta})X_2(e^{j(\omega-\theta)}) d\theta$$

- The second is a **periodic (circular) convolution** of the two spectra, scaled by $1/2\pi$, integrated over any 2π interval.
- **It has to be periodic**: both spectra are 2π -periodic, so an ordinary infinite convolution would diverge.
- Proof in one move: substitute the inverse DTFT of $x_1[n]$ into the forward DTFT of the product, then swap the sum and the integral.
- This is the windowing property used to explain Gibbs oscillation in Ch 5: truncating $h[n]$ with a window multiplies in time, so it convolves in frequency and smears the band edge.

1.8 Properties of a discrete-time system

Test the given system for linearity, time-invariance, causality, memory and BIBO stability **3 5 10**

TOP 12/45 76 Ch, 74 Ch, 82 Ba, 78 Ch, 75 Ash, 71 Bh, 70 Bh, 68 Bh, 67 Mng, 66 Ma, 69 Bh, 80 Bh · plus 81 Ba on BIBO alone · every instance worked in Ch1-num §9

The six properties, with the test that proves each

- **Static / memoryless:** $y[n]$ depends only on $x[n]$ at the same n .
 - Test: does any shifted index $x[n - k]$, $x[2n]$ or past output appear? If yes, **dynamic**.
- **Invertible:** distinct inputs give distinct outputs, so an inverse system exists with $h[n] * h_1[n] = \delta[n]$.
- **Causal:** $y[n]$ uses only present and past inputs.
 - Test: look for $x[n + k]$, $k > 0$, or a folded index. $y[n] = x[-n]$ is **non-causal**: at $n = -2$ it needs $x[2]$.
- **Time invariant:** shift the input, the output shifts identically.
- **Linear:** superposition holds.
- **BIBO stable:** every bounded input gives a bounded output.

BIBO stability

- Bounded means $|x[n]| \leq M_x < \infty$ for all n .
- **To disprove:** find one bounded input giving an unbounded output. A constant or $u[n]$ almost always works.
 - Eg $y[n] = nx[n]$ with $x[n] = 1$ gives $y[n] = n \rightarrow \infty \times$ unstable.
 - Eg $y[n] = \sum_{k=0}^n x[k]$ (accumulator) with $x[n] = u[n]$ gives $y[n] = n + 1 \times$ unstable. That is **80 Bh** in one line.
- **To prove** for an LTI system, use the impulse-response test in §1.9 instead: $\sum_k |h[k]| < \infty$.
- Recursive example, **81 Ba**: $y[n] = x[n] + e^a y[n - 1]$ has $h[n] = (e^a)^n u[n]$, so it is stable only if $|e^a| < 1$, i.e. **$a < 0$** .

Linearity, the only reliable way to write it

1. $x_1[n] \rightarrow y_1[n]$ and $x_2[n] \rightarrow y_2[n]$.
 2. Apply $x[n] = a x_1[n] + b x_2[n]$ and compute $y[n]$ from the system equation.
 3. Compare against $a y_1[n] + b y_2[n]$. Equal $\rightarrow$ linear.
- Eg $y[n] = nx[n]$: $n(ax_1 + bx_2) = a n x_1 + b n x_2 \checkmark$ linear. $y[n] = x^2[n]$: the cross term $2abx_1x_2$ survives $\times$ not linear.

Time invariance, the only reliable way to write it

1. Delayed input: put $x_1[n] = x[n - n_0]$ into the equation, call the result $y_1[n]$.
2. Delayed output: write $y[n - n_0]$ by replacing every n in $y[n]$, **including the explicit ones**.
3. Equal $\rightarrow$ time invariant.

Eg $y[n] = nx[n]$: $y_1[n] = nx[n - n_0]$ but $y[n - n_0] = (n - n_0)x[n - n_0] \times$ time varying. $y[n] = \sin(x[n]) \checkmark$ time invariant. **An explicit n multiplying the input is the classic time-varying flag**; folding $y[n] = x[-n]$ is time varying too.

Answer skeleton **5 10**

1. State the property in one line.
2. Apply its test to *this* system, showing the algebra.
3. Give the verdict with the reason attached, Eg “not time invariant, because the explicit n is not delayed”.

Marks are awarded per property, so answer all the parts asked even if one is obvious. A bare “yes/no” with no working scores nothing.

Results worth memorising

- $y[n] = x[-n]$: linear $\checkmark$, time varying $\times$, non-causal $\times$, stable $\checkmark$.
- $y[n] = x[n^2]$ or $x[2n]$: linear $\checkmark$, time varying $\times$.
- $y[n] = x^2[n]$, $e^{x[n]}$, $\log x[n]$: **non-linear** $\times$, but time invariant $\checkmark$ and memoryless $\checkmark$.
- $y[n] = y[n - 4] + x[n - 4]$: linear $\checkmark$, time invariant $\checkmark$, causal $\checkmark$, **dynamic and unstable**.
- $y[n] = \cos(\frac{5\pi}{8}n + \frac{\pi}{4})$ has no input at all: not a system response, so it fails linearity trivially.

1.9 LTI systems and the convolution sum

What is the convolution summation? Derive its equation and explain its significance 1+3 2 3

HOT 4/45 69 Bh, 73 Ma, 70 Asa, 70 Bh · the theory half of the subject's most-asked question

Derivation, four steps (**this is the full-mark answer**)

1. Decompose the input into impulses. By the sifting property,

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

i.e. $\dots + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \dots$

2. Use time invariance. If $\delta[n] \rightarrow h[n]$, then $\delta[n-k] \rightarrow h[n-k]$.

3. Use linearity. The input is a weighted sum of shifted impulses, so the output is the same weighted sum of shifted impulse responses.

4. Therefore

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

Significance, if asked 2

- $h[n]$ **alone characterises an LTI system completely**. Know the impulse response, know the output for every possible input.
- It converts a system description into an arithmetic operation, so the response can be computed rather than solved.
- Causality and stability both become simple tests on $h[n]$ (below).

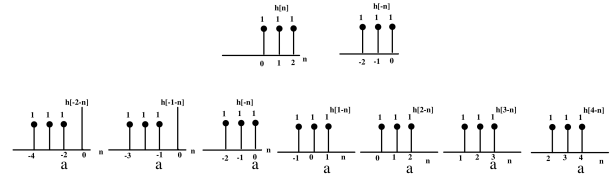
Properties of LTI systems

- **Identity**: $x[n] * \delta[n] = x[n]$.
- **Shifting**: $x[n] * \delta[n-k] = x[n-k]$.
- **Commutative**: $x[n] * h[n] = h[n] * x[n]$: fold whichever sequence is shorter.
- **Distributive**: $x * (h_1 + h_2) = x * h_1 + x * h_2 \rightarrow$ two systems in **parallel** add their impulse responses.
- **Associative**: $x * (h_1 * h_2) = (x * h_1) * h_2 \rightarrow$ two systems in **cascade** convolve their impulse responses, and their order does not matter.
- **Memoryless** $\Leftrightarrow h[n] = k \delta[n]$, giving $y[n] = kx[n]$.
- **Invertible** $\Leftrightarrow$ there is $h_1[n]$ with $h[n] * h_1[n] = \delta[n]$.
- **Causal** $\Leftrightarrow h[n] = 0$ **for** $n < 0$.
 - From $y[n] = \sum_k h[k]x[n-k]$: terms with $k < 0$ would need future inputs.
- **BIBO stable** $\Leftrightarrow \sum_k |h[k]| < \infty$ (absolutely summable).
- **Step response** $s[n] = h[n] * u[n] = \sum_{k=-\infty}^n h[k]$, so $h[n] = s[n] - s[n-1]$.

Proof of the stability condition, if asked to derive it

- $|y[n]| = |\sum_k h[k]x[n-k]| \leq \sum_k |h[k]| |x[n-k]|$ by the triangle inequality and $|ab| = |a||b|$.
- Bound the input, $|x[n-k]| \leq M_x$: $|y[n]| \leq M_x \sum_k |h[k]|$.
- So the output is bounded whenever $\sum_k |h[k]| < \infty$.

Worked cases: $h[n] = \delta[n - n_0]$ gives $\sum |h| = 1$, stable. $h[n] = u[n - n_0]$ gives $\sum |h| = \infty$, unstable. $h[n] = a^n u[n]$ gives $\sum |h| = 1/(1 - |a|)$, **stable only if** $|a| < 1$.



Building $h[a-n]$: fold $h[n]$, then slide. Each value of a gives one output sample. BB Sir, deck 3

The graphical method, four steps

1. Fold $h[k] \rightarrow h[-k]$.
2. Shift by n : $h[n-k]$.
3. Multiply with $x[k]$ sample by sample.
4. Sum the products. That single number is $y[n]$. Repeat for each n .

Shortcut for two finite sequences: the output starts at $n_x + n_h$ (sum of the starting indices) and has $L_x + L_h - 1$ samples. Use it to check your answer's length and position, every time.

Differentiate FIR and IIR systems; define recursive and non-recursive systems with examples 3+3 4

5

HOT 4/45 74 Bh, 70 Ma, 72 Ash, 80 Ba · the FIR/IIR comparison is re-used in Ch 3 and Ch 4, learn it once

FIR Vs IIR

	FIR	IIR
$h[n]$	finite, $h[n] = 0$ outside $0 \leq n \leq M - 1$	infinite length
Output Memory	$\sum_{k=0}^{M-1} h[k]x[n-k]$ finite, last M samples	$\sum_{k=0}^{\infty} h[k]x[n-k]$ infinite
Built with	convolution directly	difference equation, feedback
Stability	always stable	only if poles are inside the unit circle
Phase	can be exactly linear	cannot be exactly linear

The reason IIR needs a difference equation: convolution would need infinite memory, so it cannot be implemented directly.

Recursive Vs non-recursive

- **Non-recursive:** output from present and past *inputs* only, $y[n] = F\{x[n], \dots, x[n-M]\}$. **No feedback.** A causal FIR system is non-recursive.
 - Eg moving average
 $y[n] = \frac{1}{3}\{x[n] + x[n-1] + x[n-2]\}$.
- **Recursive:** output also uses past *outputs*, $y[n] = F\{y[n-1], \dots, y[n-N], x[n], \dots, x[n-M]\}$. **Has a feedback loop.**
 - Eg $y[n] = x[n] + a y[n-1]$.
 - Recursive form of the moving average (asked in 70 Ma): $y[n] = y[n-1] + \frac{1}{M}\{x[n] - x[n-M]\}$: one add and one subtract per sample instead of M adds.
- **FIR/IIR is about the impulse response; recursive/non-recursive is about the implementation.** They usually coincide but are not the same statement: a recursive structure can still realise an FIR response.

State the linear constant-coefficient difference equation; find the zero-input and zero-state response 4 5

HOT 5/45 80 Ba, 76 Bh, 77 Ch, 73 Ch, 81 Ba · the system-function half is in Ch 3, the response half is worked in Ch1-num §11

General form

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k], \quad a_0 = 1$$

$$\text{or } y[n] = -\sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$$

- N is the **order** of the system.
- $N = 0 \rightarrow$ non-recursive (FIR); $N \geq 1 \rightarrow$ recursive.

Total response = zero-input + zero-state

- Iterating $y[n] = a y[n-1] + x[n]$ from $y[-1]$ gives

$$y[n] = \underbrace{a^{n+1}y[-1]}_{\text{zero input}} + \underbrace{\sum_{k=0}^n a^k x[n-k]}_{\text{zero state}}$$

- **Zero-state (forced) response:** set $y[-1] = 0$. The part the input forces on the system. A system with zero initial conditions is called **relaxed**.
- **Zero-input (natural, free) response:** set $x[n] = 0$. Depends only on the system and its initial conditions.
- **An LTI system must be relaxed.** With non-zero initial conditions the system is not linear, which is why questions say “initially relaxed”.

Find the frequency response of an LTI system / its response to a complex exponential 4 5

HOT 7/45 70 Ch, 72 Ka, 80 Ba, 71 Shr, 69 Bh, 73 Shr, 82 Ba · the cheapest 5 marks in the paper, see the shortcut · worked in Ch1-num §3

- Feed $x[n] = e^{j\omega n}$ into the convolution sum:

$$y[n] = \sum_k h[k] e^{j\omega(n-k)} = e^{j\omega n} \underbrace{\sum_k h[k] e^{-j\omega k}}_{H(e^{j\omega})}$$

- So $y[n] = H(e^{j\omega}) e^{j\omega n}$: **the input comes out unchanged in shape, only scaled.**
- $e^{j\omega n}$ is therefore an **eigenfunction** of every LTI system, and $H(e^{j\omega})$, the DTFT of $h[n]$, is the **eigenvalue**, called the **frequency response**.
- $|H(e^{j\omega})|$ is the magnitude response, $\angle H(e^{j\omega})$ the phase response. Both are 2π -periodic (§1.4).

The shortcut (do not convolve these)

- The question gives $x[n] = Ae^{j\omega_0 n}$ over all n , so it is **not a finite sequence**: graphical convolution is the wrong tool.
- Compute $H(e^{j\omega})$ once. For $h[n] = a^n u[n]$:

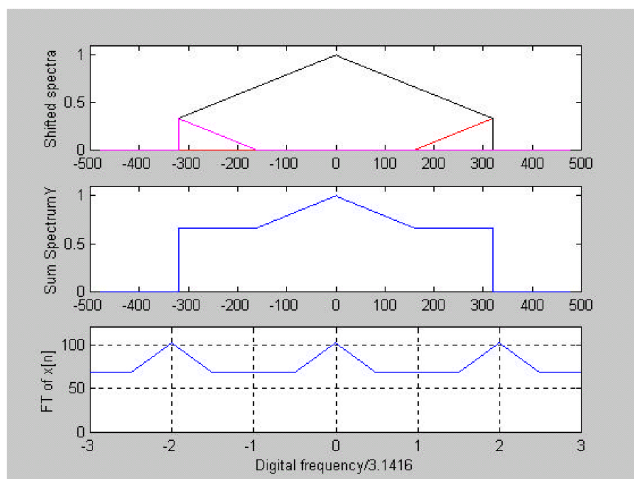
$$H(e^{j\omega}) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \frac{1}{1 - ae^{-j\omega}}$$

- Evaluate at the ω_0 the question gives.
- Answer: $y[n] = A H(e^{j\omega_0}) e^{j\omega_0 n}$.

For a sum of sinusoids (73 Shr) do this once per frequency and add: superposition. For a real cosine, write it as two exponentials at $\pm\omega_0$ and use conjugate symmetry.

1.10 Sampling of continuous-time signals

What is sampling? How is the spectrum of the sampled signal related to that of the continuous-time signal? Explain aliasing 6 8

PIN 2/45 69 Bh, 68 Bh · the numerical form, "write $x[n]$ after sampling", is asked by 7 more papers, worked in Ch1-num §10

Shifted copies, their sum, and the resulting $X(e^{j\omega})$. The copies overlap here, so the triangle is distorted: **this is aliasing**. Chapterwise Ch2 p43

The relation (quote this equation)

- Sampling means $x[n] = x_c(nT)$, with T the sampling period and $F_s = 1/T$ the sampling rate.
- Substituting into the inverse CTFT and regrouping gives

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\frac{\omega}{T} + j\frac{2\pi k}{T}\right)$$

- In words: **the spectrum of the sampled signal is the spectrum of the analog signal, scaled by $1/T$, repeated every $\Omega_s = 2\pi/T$, with the analog frequency mapped to $\omega = \Omega T$.**
- Three-step recipe for the diagram: (1) cut $X_c(j\Omega)$ into bands of width $2\pi/T$; (2) shift every band to the centre; (3) add.

Aliasing

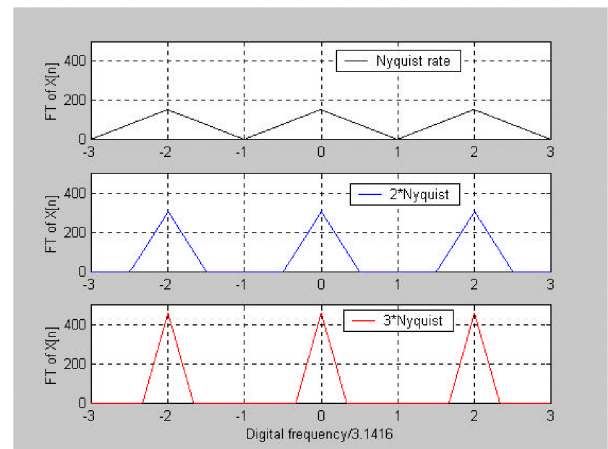
- If the copies are spaced closer than the signal is wide, they **overlap** and add. The overlapped region can no longer be separated.
- A high frequency then reappears as, and is indistinguishable from, a lower one.
- Cure**: sample faster, or band-limit the input first.

Sampling theorem

- Aliasing disappears when

$$\Omega_s = \frac{2\pi}{T} \geq 2W \quad \text{or} \quad F_s \geq 2F_{max}$$

- $2F_{max}$ is the **Nyquist rate**; $F_s/2$ is the **folding frequency**.
- **Oversampling** means F_s is taken well above the Nyquist rate: it pushes the copies further apart, so the anti-aliasing filter can be gentler (76 Bh asks exactly this).
- When the criterion holds, every copy but $k = 0$ is zero in $-\pi \leq \omega < \pi$, so $X(e^{j\omega}) = \frac{1}{T} X_c(j\omega/T)$: **the shape is preserved**, only scaled. That is why perfect reconstruction is possible.
- The relative width of the digital spectrum is $\rho = 2W/\Omega_s$: the faster you sample, the narrower the signal sits inside the 2π span.



Same signal at the Nyquist rate, twice it, three times it. The copies pull apart and the gap between them grows. Chapterwise Ch2 p45

Why is an anti-aliasing filter needed? Describe the effect of the sample-and-hold circuit at the input of an A/D converter 2+5

PIN 1/45 69 Bh · EX 753 only, but it is the theory behind every sampling numerical

Anti-aliasing filter

- A real signal is never perfectly band-limited: noise and harmonics run above $F_s/2$.
- Those components would fold down into the band of interest and **cannot be removed afterwards**: once aliased, the damage is permanent.
- So an **analog low pass filter** is placed *before* the sampler, with cut-off at or below $F_s/2$.
- It is analog by necessity: a digital filter would come after the sampling that caused the problem.
- Practical filters are not brickwall, which is the second reason to oversample: it widens the transition band the filter is allowed to use.

Sample-and-hold

- An A/D converter needs a **constant** input during its conversion time. A signal still moving would be converted to a wrong, smeared value.
- The S/H circuit tracks the input, then freezes it at the sampling instant and holds it flat until the conversion finishes.
- Effects to state:
 - Holding is a zero-order hold: it multiplies the spectrum by a sinc-shaped envelope, so high frequencies droop.
 - That droop is corrected later by a digital $\times 1/\text{sinc}$ equaliser.
 - Real S/H adds aperture jitter, droop during hold and a small pedestal error.
- **Reconstruction** at the output is the mirror image: D/A holds each sample flat, then an analog *smoothing* low pass filter removes the spectral images.

1.11 Last-minute recall for this chapter

Must memorise

- $x_e = \frac{1}{2}(x[n] + x[-n])$, $x_o = \frac{1}{2}(x[n] - x[-n])$.
- Periodic $\Leftrightarrow \omega_0/2\pi$ rational; then $N = 2\pi m/\omega_0$. **No π in ω_0 means aperiodic.**
- Energy signal: $0 < E < \infty$, $P = 0$. Power signal: $0 < P < \infty$, $E = \infty$. Periodic $\rightarrow$ power. $|e^{j\theta}| = 1 \rightarrow$ power, $P = 1$.
- E for $a^n u[n]$ is $1/(1 - a^2)$; P for $u[n]$ is $\frac{1}{2}$.
- $y[n] = \sum_k x[k]h[n - k]$. Output starts at $n_x + n_h$, length $L_x + L_h - 1$.
- LTI causal $\Leftrightarrow h[n] = 0$, $n < 0$. LTI stable $\Leftrightarrow \sum |h[k]| < \infty$.
- $e^{j\omega n}$ in $\rightarrow H(e^{j\omega})e^{j\omega n}$ out. Never convolve these.
- $X(e^{j\omega}) = \frac{1}{T} \sum_k X_c(j\frac{\omega + 2\pi k}{T})$, and $F_s \geq 2F_{max}$.

Most repeated, in order

1. Convolution, find $y[n]$ **TOP 34/45**
2. Energy Vs power + periodicity **TOP 10/45**
3. Test a system for linearity / TI / causality / stability **TOP 12/45**
4. Find the fundamental period **HOT 7/45**
5. Response to a complex exponential **HOT 7/45**
6. Sample an analog signal, write $x[n]$ **HOT 7/45**
7. Convolution from tabulated or stem-plot data **HOT 4/45**
8. Even and odd parts **HOT 4/45**
9. FIR Vs IIR, recursive Vs non-recursive **HOT 4/45**
10. Convolution sum, derive and explain **HOT 4/45**

Q1 and Q2 of the paper come from this chapter in almost every year. If time is short, do §1.9 and §1.3 and leave §1.6: the DTFS has been asked 5 times in 45 papers and never for more than 6 marks.

2. Z-Transform

4 hrs · 61 questions across the 45 papers · worth about 7 of 80 marks · 3 syllabus sub-topics · the shortest chapter, and the best marks-per-hour in the subject.

Read this first:

- Only **two** things are ever asked, and they are asked *together*:
 - state the properties of the ROC **TOP 14/45**: a 1 to 5 mark bookwork question, **same answer every year**,
 - find the inverse Z-transform of a given $X(z)$ with a given ROC **TOP 32/45**, for 5 or 6 marks.
- Those two appear as **one** question, Q3, in almost every paper. Learn §2.2 and §2.5 and the chapter is finished.
- The ROC is not decoration.** The same $X(z)$ has a different $x[n]$ for every ROC, and papers exploit that: 81 Ba, 82 Ba and 71 Ch each ask for **three** answers from one $X(z)$.
- Properties with proofs (convolution **HOT 5/45**, time shift, scaling) are worth 3 to 6 marks and come from §2.3.
- Use one method for every inverse:** divide by z , partial-fraction $X(z)/z$, multiply back so every term reads $z/(z-a)$. §2.5 explains why.

2.1 Definition, and how it relates to the DTFT

Define the Z-transform. How is it related to the DTFT? Explain the scaling property **3+3 2**

PIN 3/45 74 Bh, 76 Bh, 73 Ma · the definition is the opening line of most Ch 2 answers

Bilateral (two-sided) Z-transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}, \quad z = r e^{j\omega}$$

- z is a **complex** variable, so $X(z)$ is defined on a plane, not a line.
- The sum exists only where it converges: that region is the **ROC**, and $X(z)$ **without its ROC is only half an answer**.
- Splitting the sum shows where the two halves live:

$$X(z) = \underbrace{\sum_{n=-\infty}^{-1} x[n] z^{-n}}_{\text{left-sided part}} + \underbrace{\sum_{n=0}^{\infty} x[n] z^{-n}}_{\text{right-sided part}}$$

The right half needs $|z|$ *large*, the left half needs $|z|$ *small*; an ROC is where both hold at once.

Unilateral (one-sided) Z-transform

$$X^+(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

- Ignores $n < 0$ entirely, so its ROC is always the outside of a circle and **need not be stated**.
- Importance:** it is the only version that can carry **initial conditions**, so it is what solves a difference equation for a system that is not relaxed.

Standard pairs (**memorise these five**)

Relation to the DTFT (**quote this**)

$$X(e^{j\omega}) = X(z) \Big|_{z=e^{j\omega}}$$

- Put $r = 1$: the Z-transform evaluated **on the unit circle** is the DTFT.
- So the DTFT exists **if and only if the ROC contains the unit circle**. That one sentence answers half the theory questions in this chapter.
- In reverse: $X(z)$ is the DTFT of the exponentially weighted sequence $x[n] r^{-n}$. The r^{-n} is what lets the Z-transform handle growing sequences the DTFT cannot.
- Going round the unit circle once is ω advancing by 2π : the same 2π -periodicity as §1.4.

- Time delay property** of the one-sided transform (73 Ma asks this):

$$Z^+\{x[n-k]\} = z^{-k} X^+(z) + \sum_{m=1}^k x[-m] z^{-(k-m)}$$

- The extra sum is the initial conditions**; the bilateral version has no such term, which is exactly why the one-sided transform is the one that solves a difference equation with given $y[-1]$, $y[-2]$.

$x[n]$	$X(z)$	ROC	Note
$\delta[n]$	1	all z	finite duration
$u[n]$	$\frac{1}{1 - z^{-1}} = \frac{z}{z - 1}$	$ z > 1$	pole on the unit circle, so not stable
$a^n u[n]$	$\frac{1}{1 - az^{-1}} = \frac{z}{z - a}$	$ z > a $	right-sided, causal
$-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}} = \frac{z}{z - a}$	$ z < a $	left-sided, anticausal
$n a^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $	repeated pole; also $\frac{az}{(z - a)^2}$

Rows 3 and 4 have the identical $X(z)$ and differ only in the ROC. That is the whole reason the ROC must be given, and the reason an inverse cannot be started until you have read it.

2.2 Region of convergence

Define the ROC. State and explain the properties of the region of convergence 1 2 3 4 5

TOP 14/45 80 Bh, 73 Ch, 72 Ch, 81 Bh, 77 Ch, 71 Bh, 70 Bh, 79 Ba, 72 Ka, 71 Shr, 73 Bh, 78 Ch, 70 Ma, 74 Ma · pure bookwork, and it is asked every single year

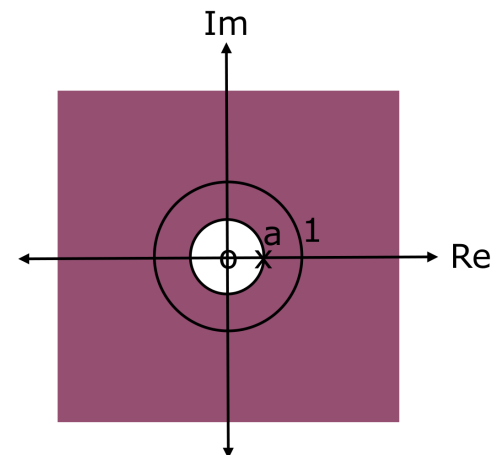
Definition

- The ROC is the set of values of z for which $\sum_n |x[n]| |z|^{-n}$ converges, i.e. where $X(z)$ actually exists.
- It depends only on $|z| = r$, never on the angle, which is why it is always drawn as circles.

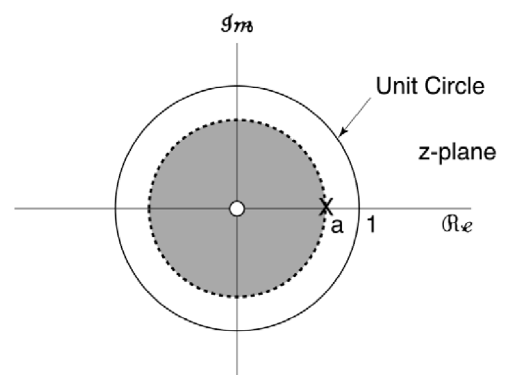
The eight properties (this is the answer, in this order)

- The ROC is a **ring or a disc** in the z -plane, centred on the origin.
- The DTFT of $x[n]$ converges absolutely if and only if the ROC includes the unit circle.
- The ROC cannot contain any pole. $X(z)$ is infinite at a pole, so the sum cannot converge there.
- If $x[n]$ is **finite duration**, the ROC is the entire z -plane, possibly except $z = 0$ and/or $z = \infty$.
- If $x[n]$ is **right-sided** ($x[n] = 0$ for $n < N_1$), the ROC extends **outward** from the outermost finite pole, possibly including $z = \infty$.
- If $x[n]$ is **left-sided** ($x[n] = 0$ for $n > N_2$), the ROC extends **inward** from the innermost non-zero pole, possibly including $z = 0$.
- If $x[n]$ is **two-sided**, the ROC is a **ring** bounded inside and outside by a pole, containing none.
- The ROC must be a **connected region**.

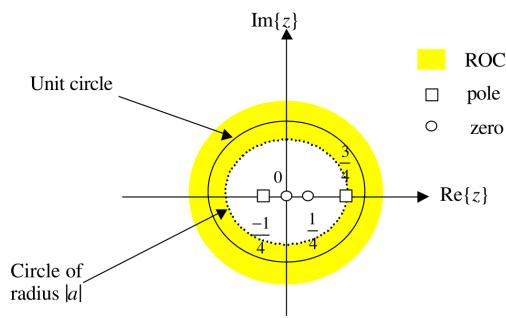
For a 4 or 5 mark answer add one line of *why* to properties 3, 5 and 6, and sketch the three cases. For 1 or 2 marks, the definition plus properties 1, 3 and 5 is enough.



Right-sided $x[n] = a^n u[n]$: the series $\sum (az^{-1})^n$ converges for $|az^{-1}| < 1$, i.e. $|z| > |a|$: everything *outside* the pole circle. BB Sir, deck 2.1



Left-sided $x[n] = -a^n u[-n - 1]$: same $X(z) = z/(z - a)$, but the ROC is $|z| < |a|$, the disc *inside* the pole circle. BB Sir, deck 2.1



Two-sided: the ROC is the ring between two poles. It contains the unit circle here, so this signal has a DTFT. Chapterwise Ch3

Stability and causality from the ROC (asked with the properties)

- **Causal** $\Leftrightarrow h[n] = 0$ for $n < 0 \Leftrightarrow$ the ROC is the **outside of the outermost pole**, and includes $z = \infty$.
- **BIBO stable** $\Leftrightarrow \sum |h[n]| < \infty \Leftrightarrow$ the **ROC contains the unit circle**.
- **Causal and stable together** $\Leftrightarrow$ **every pole lies strictly inside the unit circle**. This is the single most quoted line in Chapters 2, 3 and 6.
- A stable but non-causal system is possible: the ROC is a ring that contains the unit circle but does not reach $z = \infty$.

Reading the ROC off a question (the skill the numericals need)

1. Find the poles.
2. Mark their radii on a line.
3. The given ROC lies in one of the gaps. Any pole with radius below that gap is **right-sided**; any pole above it is **left-sided**.

A given ROC that contains a pole is impossible: two papers print exactly that, see Ch2-num §4.

2.3 Properties of the Z-transform

State and prove the convolution property; describe causality and stability in terms of the ROC 2+4

3+3 1+3+3

HOT 5/45 76 Ash, 72 Ka, 70 Asa, 72 Ma, 72 Ash · **the most-asked property by far**

Property	Transform	ROC
Linearity	$a x_1[n] + b x_2[n] \rightarrow a X_1(z) + b X_2(z)$	at least $R_{x_1} \cap R_{x_2}$
Time shift	$x[n - k] \rightarrow z^{-k} X(z)$	R_x , except possibly $z = 0$ or $z = \infty$
Scaling in z (multiply by a^n)	$a^n x[n] \rightarrow X(a^{-1}z)$	$ a r_1 < z < a r_2$
Differentiation	$n x[n] \rightarrow -z \frac{dX(z)}{dz}$	R_x
Time reversal	$x[-n] \rightarrow X(z^{-1})$	$\frac{1}{r_2} < z < \frac{1}{r_1}$
Convolution	$x_1[n] * x_2[n] \rightarrow X_1(z) X_2(z)$	at least $R_{x_1} \cap R_{x_2}$
Multiplication	$x_1[n] x_2[n] \rightarrow \frac{1}{2\pi j} \oint_C X_1(v) X_2\left(\frac{z}{v}\right) v^{-1} dv$	at least $r_{1_1} r_{1_2} < z < r_{2_1} r_{2_2}$

Convolution, proved (the 3 to 6 mark answer)

- Start from the definition of the convolution sum:
 $x[n] = \sum_k x_1[k] x_2[n - k]$.
- Transform both sides:

$$X(z) = \sum_n \left[\sum_k x_1[k] x_2[n - k] \right] z^{-n}$$

- Interchange the order of summation** (the step that earns the mark):

$$X(z) = \sum_k x_1[k] \sum_n x_2[n - k] z^{-n}$$

- The inner sum is the **time-shift property**:
 $\sum_n x_2[n - k] z^{-n} = z^{-k} X_2(z)$.

$$X(z) = X_2(z) \sum_k x_1[k] z^{-k} = X_1(z) X_2(z)$$

- “At least” the intersection:** a pole of one can be cancelled by a zero of the other, which makes the ROC larger than the intersection. Say this: it is the part most answers miss.

Time shift, proved 3 69 Ch

- Let $y[n] = x[n - k]$, so $Y(z) = \sum_n x[n - k] z^{-n}$.
- Substitute $m = n - k$, so $n = m + k$:

$$Y(z) = \sum_m x[m] z^{-(m+k)} = z^{-k} \sum_m x[m] z^{-m} = z^{-k} X(z)$$

- Why the ROC can change:** z^{-k} adds poles at $z = 0$ (for $k > 0$) or at $z = \infty$ (for $k < 0$), so those two points may drop out.

Scaling, proved 3 74 Bh

$$Z\{a^n x[n]\} = \sum_n a^n x[n] z^{-n} = \sum_n x[n] (a^{-1} z)^{-n} = X(a^{-1} z)$$

- The ROC $r_1 < |z| < r_2$ becomes $r_1 < |a^{-1} z| < r_2$, i.e. $|a| r_1 < |z| < |a| r_2$: **every pole is scaled by a** .
- Example: $u[n] \rightarrow \frac{1}{1-z^{-1}}$, so $a^n u[n] \rightarrow \frac{1}{1-az^{-1}}$, pole moved from 1 to a .

Differentiation, used in every na^n question

- Differentiate $X(z) = \sum x[n] z^{-n}$ term by term and multiply by $-z$:

$$Z\{n x[n]\} = -z \frac{dX(z)}{dz}$$

- Applied to $a^n u[n]$ it gives $Z\{na^n u[n]\} = \frac{az^{-1}}{(1-az^{-1})^2}$, $|z| > |a|$: **the answer 78 Ch and 70 Ma both ask for**.

2.4 Finding $X(z)$ and locating the ROC**Find the Z-transform of the given signal and locate its ROC 2+4 3+6 4+6**

HOT 6/45 70 Ch, 70 Asa, 75 Ash, 75 Ch, 75 Bh, 73 Ma · worked in Ch2-num §6

Method

- Split $x[n]$ into its terms.
- Transform each with a standard pair, **writing its own ROC beside it**.
 - $a^n u[n] \rightarrow \frac{1}{1-az^{-1}}$, ROC $|z| > |a|$ (outside).
 - $-b^n u[-n-1] \rightarrow \frac{1}{1-bz^{-1}}$, ROC $|z| < |b|$ (inside).
 - Watch the sign:** a $+b^n u[-n-1]$ term transforms to $-\frac{1}{1-bz^{-1}}$.
- The ROC of the sum is the **intersection** of the individual ROCs.
- If the intersection is empty, **the Z-transform does not exist**: and that is the full answer. 70 Ch prints exactly such a signal (Ch2-num §6).

Two standard transforms the papers ask for by name

Why the intersection can be empty

- A right-sided term needs $|z| > R_1$; a left-sided term needs $|z| < R_2$.
- The ring $R_1 < |z| < R_2$ exists **only if $R_1 < R_2$** : the right-sided pole must be the *inner* one.
- $R_1 = R_2$ or $R_1 > R_2$ leaves nothing, so $X(z)$ does not exist even though each term separately transforms perfectly well.

- $x[n] = na^n u[n]$ (78 Ch, 70 Ma): use differentiation:

$$X(z) = \frac{az^{-1}}{(1 - az^{-1})^2}, \quad |z| > |a|$$

- Derivation in one line:

$$-z \frac{d}{dz} \frac{z}{z-a} = -z \cdot \frac{(z-a) - z}{(z-a)^2} = \frac{az}{(z-a)^2}.$$

- $x[n] = \cos \omega n$ for $n \geq 0$ (74 Ma): write the cosine as two exponentials and add two geometric series:

$$X(z) = \frac{1 - z^{-1} \cos \omega}{1 - 2z^{-1} \cos \omega + z^{-2}}, \quad |z| > 1$$

Its two poles are $e^{\pm j\omega}$, on the unit circle, which is why the ROC excludes it and why a pure sinusoid is not BIBO stable.

2.5 Inverse Z-transform: the three methods

Determine the inverse Z-transform of the given $X(z)$ for the stated ROC 5 6 1+5 2+6

TOP 32/45 · the single biggest question in this chapter, 5 or 6 marks in nearly every paper · every instance is worked in Ch2-num

Which method, and when

1. **Partial fractions:** the default. Use unless the question names another method. Gives a closed form for every n .
2. **Long division (power series):** use when the question says so (77 Ch, 76 Bh, 73 Bh, 72 Ma, 70 Ma). Gives numbers, **not a closed form**, so give 4 or 5 terms and the recurrence.
3. **By inspection:** when $X(z)$ is already a finite polynomial, read the coefficients straight off as impulses (74 Ash, 78 Ch).
4. The contour-integral / residue method is never asked in these 45 papers.

The partial-fraction method, in one fixed order

1. Is $X(z)$ proper? If the numerator's degree is $\geq$ the denominator's, **long-divide first** and keep the polynomial part aside: it becomes impulses.
2. Divide by z and expand $X(z)/z$ in partial fractions.
3. Multiply back by z , so every term reads $\frac{Az}{z-a}$.
4. Assign each pole a side from the ROC (§2.2).
5. Write the answer term by term with the pairs opposite.

Long division: which way to divide (where the marks are lost)

- **Right-sided** (ROC outside, causal): arrange both polynomials in **ascending powers of z^{-1}** and divide. The quotient $x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots$ is read off directly.
- **Left-sided** (ROC inside, anticausal): arrange in **ascending powers of z** instead. The quotient gives $x[-1]z, x[-2]z^2, \dots$
- **Dividing the wrong way round gives the other sequence**, which is a valid series for a different ROC: and scores zero.

Why divide by z first

- The standard pairs are $\frac{z}{z-a} \leftrightarrow a^n u[n]$, with a z on top.
- Expanding $X(z)$ directly gives terms $\frac{A}{z-a}$ with no z on top, and each then needs an extra time shift. Expanding $X(z)/z$ puts the z back for free.
- **Use it every single time**, including the repeated-pole cases. One idiom, no decisions.

The pair table, both sides

Term	ROC $ z > a $	ROC $ z < a $
$\frac{z}{z-a}$	$a^n u[n]$	$-a^n u[-n-1]$
$\frac{z}{(z-a)^2}$	$n a^{n-1} u[n]$	$-n a^{n-1} u[-n-1]$
constant C	$C\delta[n]$	$C\delta[n]$
z^k	$\delta[n+k]$, an <i>advance</i>	

The right-hand column is just the left with a minus sign and $u[-n-1]$ instead of $u[n]$. That is the only thing the ROC changes.

- Faster than dividing: read the **recurrence** straight off the denominator. For $X(z) = \frac{1}{1 - a_1 z^{-1} - a_2 z^{-2}}$,

$$x[n] = a_1 x[n-1] + a_2 x[n-2], \quad x[0] = 1$$

and then just iterate. This is the same difference equation the filter implements (Ch 4).

- **Always give the recurrence as well as the terms:** it shows the pattern continues and it is what a 6-mark answer wants.

2.6 Last-minute recall for this chapter

Must memorise

- $X(z) = \sum_n x[n]z^{-n}$; the DTFT is $X(z)$ on the unit circle.
- $a^n u[n] \leftrightarrow \frac{z}{z-a}, |z| > |a|$; $-a^n u[-n-1] \leftrightarrow \frac{z}{z-a}, |z| < |a|$.
Same $X(z)$, different ROC.
- $na^n u[n] \leftrightarrow \frac{az^{-1}}{(1-az^{-1})^2}$; $\frac{z}{(z-a)^2} \leftrightarrow na^{n-1}u[n]$.
- The 8 ROC properties (§2.2): **written out, in order.**
- Causal $\Rightarrow$ ROC outside the outermost pole. Stable $\Rightarrow$ ROC contains the unit circle. **Both $\Rightarrow$ all poles inside the unit circle.**
- Convolution $\rightarrow$ product; shift $\rightarrow z^{-k}$; multiply by $a^n \rightarrow X(a^{-1}z)$; multiply by $n \rightarrow -z dX/dz$.
- Inverse: proper? $\rightarrow$ divide by $z \rightarrow$ partial fractions $\rightarrow$ assign sides from the ROC.

Most repeated, in order

1. Inverse Z-transform with a given ROC **TOP 32/45**
2. Properties of the ROC **TOP 14/45**
3. Find $X(z)$ and locate its ROC **HOT 6/45**
4. Convolution property, stated and proved **HOT 5/45**
5. Z-transform of $na^n u[n]$ **PIN 3/45**
6. One-sided Z-transform and its time-delay property **PIN 2/45**
7. Time-shift property, derived **PIN 1/45**
8. Scaling property with an example **PIN 1/45**

Q3 of the paper is the ROC properties plus an inverse Z-transform in 30 of the 45 papers. If you prepare one question in this subject, prepare that one.

3. Analysis of LTI Systems in the Frequency Domain

6 hrs · **56 questions across the 45 papers** · worth about **8.4 of 80 marks** · 2 syllabus sub-topics · **6th of the 8 chapters** by weight: only Ch 2 and the EX 753 extras are lighter.

Read this first:

- **35 of the 56 questions are one question**, asked again and again with the numbers changed: *plot the poles and zeros in the z -plane, then sketch the magnitude response, not to scale* **TOP 35/45**. It is worth 6 to 10 marks and it is the single most valuable thing in this chapter.
- It arrives in two disguises. Either a **difference equation** is given and you must find $H(z)$ first, or the **poles and zeros are handed to you** directly, sometimes in polar form. The second half of the work is identical.
- The remaining 21 questions are short bookwork worth 2 to 4 marks each: causality and stability from the impulse response and the ROC **HOT 7/45**, the LCCDE and its system function **HOT 6/45**, FIR against IIR **PIN 3/45**, linear phase **PIN 3/45**, zero-input and zero-state response **PIN 2/45**.
- **Sketching does not mean guessing**. Three numbers, $|H|$ at $\omega = 0, \pi/2$ and π , plus the pole and zero angles, fix the shape completely. §3.3 gives the recipe.
- The cheapest check in the chapter: $H(e^{j0}) = H(1)$ is the sum of all coefficients and $H(e^{j\pi}) = H(-1)$ is their alternating sum. Both are one line of arithmetic, and both are marks.

3.1 Frequency response of an LTI system

Derive the expression for the frequency response of an LTI system. Show that a complex exponential is an eigenfunction of an LTI system **6**

PIN 2/45 69 Bh, 67 Mng · the opening paragraph of most answers in this chapter

The eigenfunction property

Feed a complex exponential into an LTI system with impulse response $h[n]$ and convolve:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)} = e^{j\omega n} \underbrace{\sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}}_{H(e^{j\omega})}$$

- The input comes out unchanged in *shape*, scaled by a complex number. So $e^{j\omega n}$ is an **eigenfunction** of every LTI system and $H(e^{j\omega})$ is the **eigenvalue**.
- $H(e^{j\omega})$ is the **frequency response**: the DTFT of $h[n]$, and equally $H(z)$ evaluated on the unit circle.

$$H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}}$$

- **This substitution is legal only if the ROC contains the unit circle**. If it does not, the system has no frequency response at all, however neatly $H(z)$ factors. See §3.4.

Magnitude, phase and group delay

- Writing $H(e^{j\omega}) = |H(e^{j\omega})| e^{j \arg H(e^{j\omega})}$, the three standard plots are the **magnitude response** $|H|$, the **phase response** $\arg H$, and the **group delay**

$$\text{grd } H(e^{j\omega}) = -\frac{d}{d\omega} \arg H(e^{j\omega})$$

- Group delay is the delay the system applies to the envelope of a narrow band of frequencies. **Constant group delay means linear phase means no phase distortion**: see §3.6.

Response to a real sinusoid

For $x[n] = A \cos(\omega_0 n + \phi)$ and a real $h[n]$, superposing the two exponentials gives

$$y[n] = A |H(e^{j\omega_0})| \cos(\omega_0 n + \phi + \angle H(e^{j\omega_0}))$$

- The system can only do two things to a sinusoid: **scale it** by $|H|$ and **shift it** by $\angle H$. It can never change its frequency.
- That is the whole reason the magnitude and phase responses are worth plotting.

Three properties that halve the work

- $H(e^{j\omega})$ is **periodic in ω with period 2π** , because $e^{j\omega n}$ is.
- For real $h[n]$, $|H|$ is **even** and $\angle H$ is **odd** in ω .
- So **everything is known from $0 \leq \omega \leq \pi$** , and that is the only range any paper asks you to draw.

- Log magnitude is quoted in dB as $20 \log_{10} |H(e^{j\omega})|$.

3.2 The LCCDE and its system function

State the linear constant coefficient difference equation and the corresponding system function. Why do we need a difference equation? **3 2**

HOT 6/45 73 Shr **3**, 71 Shr **3+2**, 69 Ch **2**, 69 Bh **2**, 73 Ch **2** · pure bookwork, **same answer every year**

The equation

An N th-order linear constant-coefficient difference equation relates the output samples to the input samples with coefficients that do not change with n :

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

- **Linear:** no products or powers of x or y , no $\sin y[n]$.
- **Constant coefficient:** a_k and b_k are numbers, not functions of n . This is what makes the system time-invariant.
- **Order N :** how many past outputs are fed back.
- Normalising $a_0 = 1$ gives the recursive computing form, which is how it is actually run on hardware:

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k]$$

The system function

Z-transform both sides, using linearity and the time-shift property $y[n-k] \leftrightarrow z^{-k}Y(z)$:

$$Y(z) \sum_{k=0}^N a_k z^{-k} = X(z) \sum_{k=0}^M b_k z^{-k}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \frac{B(z)}{A(z)}$$

- **An LCCDE always gives a rational $H(z)$** , and every rational $H(z)$ comes from some LCCDE. The two descriptions are the same object.
- **Zeros** come from the x side (feedforward), **poles** from the y side (feedback). Read them straight off the two coefficient lists.
- Multiply top and bottom by z^N to clear the negative powers before factoring.

Why we need a difference equation (a 2 mark question, asked five times)

- It is a **complete description** of an LTI system in the time domain, finite and exact, whereas $h[n]$ may be an infinitely long list.
- It is **recursive**: each output uses a handful of stored past values, so an IIR response of infinite length costs finite memory and finite arithmetic.
- It maps **directly onto hardware**: every term is one delay, one multiplier and one adder, which is exactly what a realisation structure draws (Ch 4).
- It gives $H(z)$ **by inspection**, so the frequency response, the poles, the zeros and the stability all follow from the same few coefficients.
- It carries **initial conditions**, which is what separates the zero-input from the zero-state response.

A degree count worth remembering: if $M < N$, then $B(z)$ written in positive powers is short by $N - M$, so $H(z)$ **has $N - M$ zeros at the origin**. They are real zeros and they belong on the plot, but being at $z = 0$ they are equidistant from every point of the unit circle and so **contribute nothing to the magnitude shape**.

3.3 Pole-zero plot to magnitude response

Plot the poles and zeros in the z -plane and sketch the magnitude response (not to scale) of the system described by the given difference equation **3+7 2+8 6**

TOP 35/45 80 Bh, 81 Ba, 76 Ash, 79 Bh, 79 Ba, 78 Bh, 81 Bh, 80 Ch, 72 Ash, 74 Ash, 72 Ka, 72 Ch, 82 Ba, 70 Ch, 71 Bh, 73 Ma, 69 Ch, 76 Bh, 75 Ch, 69 Bh, 67 Mng, 68 Bh, 70 Asa, 70 Ma, 80 Ba, 82 Bh, 76 Ch, 75 Ash, 81 Ch, 73 Bh, 74 Ch, 75 Bh, 73 Ch, 71 Ch · **worked in full in the numerical companion, §1 and §2**

The geometric idea, which is the whole method

Factor $H(z)$ into its zeros z_k and poles p_k :

$$H(z) = K \frac{\prod_{k=1}^M (z - z_k)}{\prod_{k=1}^N (z - p_k)}$$

Now put $z = e^{j\omega}$, a point walking round the unit circle. Taking magnitudes turns the product into distances:

$$|H(e^{j\omega})| = |K| \frac{\prod_k |e^{j\omega} - z_k|}{\prod_k |e^{j\omega} - p_k|}$$

- Every factor is the **length of the vector** drawn from a zero or a pole to the point $e^{j\omega}$.
- **Zero distances multiply on top, pole distances divide underneath.**
- So as ω sweeps 0 to π : passing **near a pole** makes a denominator distance small, and $|H|$ **peaks**; passing **near a zero** makes a numerator distance small, and $|H|$ **dips**.
- A zero exactly on the unit circle gives a distance of zero, so $|H|$ is **exactly zero** there: a perfect null.
- The closer a pole sits to the circle, the taller and narrower its peak.

The algebraic form, for when a paper wants the expression and not only the picture

Each first-order factor, written with the pole or zero at radius r and angle θ , contributes

$$|1 - re^{j\theta}e^{-j\omega}|^2 = 1 + r^2 - 2r \cos(\omega - \theta)$$

- In dB: $20 \log_{10} |1 - re^{j\theta}e^{-j\omega}| = 10 \log_{10} (1 + r^2 - 2r \cos(\omega - \theta))$, **added** for a zero factor and **subtracted** for a pole factor.
- Phase: $\arg(1 - re^{j\theta}e^{-j\omega}) = \arctan \frac{r \sin(\omega - \theta)}{1 - r \cos(\omega - \theta)}$, again + for a zero and – for a pole.
- Group delay of the same factor: $\frac{r^2 - r \cos(\omega - \theta)}{1 + r^2 - 2r \cos(\omega - \theta)}$.
- **The minimum of $1 + r^2 - 2r \cos(\omega - \theta)$ is at $\omega = \theta$** , where it equals $(1 - r)^2$. That is why a pole at angle θ puts the peak of $|H|$ at $\omega = \theta$, and why $r \rightarrow 1$ makes it sharp.

Poles and zeros given in polar form

Several papers, and the local worked example the family is copied from, specify a radius and an angle instead of a complex number. Convert first:

$$z = re^{j\theta} = r \cos \theta + jr \sin \theta$$

For example radius 0.9 at angle 1.0376 rad gives $0.45 + j0.77$, and radius 0.892 at 2.5158 rad gives $-0.728 + j0.522$. **Those are the exact numbers the papers reuse**, so recognising them saves a conversion.

The exam recipe

1. From the LCCDE write $H(z) = B(z)/A(z)$ in z^{-1} , then multiply top and bottom by z^N so both are polynomials in z .
2. **Factor.** Roots of the numerator are zeros, roots of the denominator are poles. Use the quadratic formula: these are almost always second order and the roots almost always come out complex conjugate.
3. **Plot** the unit circle, mark each zero $\circ$ and each pole $\times$, and do not forget the zeros at the origin.
4. **Evaluate three points**, which is all a not-to-scale sketch needs:

$$|H(e^{j0})| = \left| \frac{B(1)}{A(1)} \right|, \quad |H(e^{j\pi/2})| = \left| \frac{B(j)}{A(j)} \right|, \quad |H(e^{j\pi})| = \left| \frac{B(-1)}{A(-1)} \right|$$

5. **Locate the peak** at the angle of the pole, and the dip at the angle of the zero. Join the points smoothly.
6. State the character in words: low-pass, high-pass, band-pass or notch.

$B(1)/A(1)$ is just the sum of the coefficients, and $B(-1)/A(-1)$ the alternating sum. **Two lines of arithmetic for two anchor points of the sketch.**

3.4 Causality and stability, from $h[n]$ and from the ROC

Describe the stability and causality characteristics of an LTI system in terms of its impulse response and the ROC of its transfer function, with suitable examples 4+3 4 2 3

HOT 7/45 76 Ash 4+3, 72 Ch 4, 81 Ch 2, 80 Ch 2, 73 Ma 2, 73 Bh 3, 77 Ch 3 · learn the four-line table and the one-line rule

In terms of the impulse response

Property	Condition on $h[n]$
Causal	$h[n] = 0$ for all $n < 0$
Stable (BIBO)	$\sum_{n=-\infty}^{\infty} h[n] < \infty$
– Causality says the output cannot anticipate the input: it is a statement about <i>when</i> $h[n]$ is allowed to be non-zero. – Stability says a bounded input always gives a bounded output, and the necessary and sufficient test is that $h[n]$ be absolutely summable. – The two are independent. A system can be either, both or neither.	

In terms of the ROC

Property	Condition on the ROC
Causal	exterior of the outermost pole, and includes $z = \infty$
Stable	the ROC contains the unit circle
Both	all poles strictly inside $ z = 1$
– The last line is the one to quote: a causal system is stable if and only if every pole has $p < 1$. – A pole exactly <i>on</i> the circle is marginally stable, which is not BIBO stable: the frequency response blows up there. – Stability is also why the frequency response may not exist. If the ROC misses the unit circle, $H(e^{j\omega})$ is undefined, so no magnitude response can be drawn at all.	

One $H(z)$, three systems (the standard worked example, and a favourite trap)

Take $H(z)$ with poles at $z = \frac{1}{2}$ and $z = 3$. The algebra is identical in all three cases; only the ROC differs, and the ROC is a separate piece of information the question must supply.

ROC	$h[n]$	Causal?	Stable?
$\frac{1}{2} < z < 3$	$(\frac{1}{2})^n u[n] - 2(3)^n u[-n - 1]$	no	yes
$ z > 3$	$(\frac{1}{2})^n u[n] + 2(3)^n u[n]$	yes	no
$ z < \frac{1}{2}$	$(\frac{1}{2})^n u[-n - 1] + 2(3)^n u[-n - 1]$	no	no

The middle row is the point: **a causal system with a pole outside the unit circle is unstable**, no matter how ordinary $H(z)$ looks. The top row is the only stable reading, and it is not causal.

3.5 FIR against IIR

Differentiate between an FIR system and an IIR system 4

PIN 3/45 80 Ba 4, 76 Bh 4, 67 Mng · a table answer, four or five rows is enough

	FIR	IIR
Impulse response	finite length, $h[n] = 0$ outside $0 \leq n \leq M - 1$	infinite length, decays but never ends
Difference equation	non-recursive, output from inputs only	recursive, past outputs fed back
$H(z)$	all zeros, poles only at the origin	has poles away from the origin
Stability	always stable , a finite sum is always bounded	stable only if every pole is inside $ z = 1$
Linear phase	easy , exact, by making $h[n]$ symmetric	not achievable exactly
Order needed	high, for a sharp response	low, for the same sharpness
Memory, arithmetic	more	less

- The two rows examiners look for are **FIR is always stable** and **FIR can have exactly linear phase**. Those are the reasons to pay for its higher order.
- An FIR system's $H(z) = b_0 + b_1 z^{-1} + \dots + b_{M-1} z^{-(M-1)}$ written over z^{M-1} has all its poles at $z = 0$, which is inside the unit circle: the stability claim needs no further argument.

3.6 Linear phase

What is the condition satisfied by a linear phase FIR filter? Derive the frequency response of a symmetric linear phase filter of length M , M odd 2+4 6

PIN 3/45 70 Asa 2+4, 67 Mng 6, 69 Bh 6 · the only chapter source is **FIR-design material**: this topic is taught again in Ch 5

What linear phase means, and why it is wanted

- Phase is **linear** when $\arg H(e^{j\omega}) = -\alpha\omega$, a straight line through the origin.
- Then the group delay $-\frac{d}{d\omega} \arg H = \alpha$ is **constant**: every frequency is held up by the same α samples.
- A common delay shifts the signal but does not deform it. **Linear phase means no phase distortion**, which matters wherever shape carries the information: speech, ECG, data pulses, images.
- If the phase were curved, different frequency components would arrive with different delays and a pulse would spread out, even with a perfectly flat $|H|$.
- **Generalized** linear phase allows a constant offset, $\arg H = \beta - \alpha\omega$. The group delay is still constant, so it still does not distort; $\beta = \pm\pi/2$ is what an antisymmetric $h[n]$ produces.

The condition

An FIR filter of length M has linear phase if its impulse response is symmetric or antisymmetric about its midpoint:

$$h[n] = \pm h[M-1-n], \quad 0 \leq n \leq M-1$$

- + gives **symmetric** $h[n]$ and phase $-\alpha\omega$ with $\alpha = \frac{M-1}{2}$.
- – gives **antisymmetric** $h[n]$ and phase $\frac{\pi}{2} - \alpha\omega$, with the same α .
- $\alpha = (M-1)/2$ is **the centre of the sequence**, and it is the delay in samples. For odd M it is an integer; for even M it is a half-integer, so the filter delays by half a sample.
- In the z -plane the condition forces zeros to occur in conjugate reciprocal groups: if z_0 is a zero, so are z_0^* , $1/z_0$ and $1/z_0^*$.

Derivation for symmetric $h[n]$ with M odd (the 6 mark answer)

- Start from the definition and pull out the centre term at $n = \alpha = (M-1)/2$:

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h[n] e^{-j\omega n}$$

- Substitute $n \rightarrow M-1-n$ in the sum and use $h[n] = h[M-1-n]$ to pair each term with its mirror image. Each pair contributes

$$h[n](e^{-j\omega n} + e^{-j\omega(M-1-n)}) = 2h[n] e^{-j\omega\alpha} \cos(\omega(\alpha - n))$$

- Summing the α pairs and the lone centre term factors the whole response:

$$H(e^{j\omega}) = e^{-j\omega\alpha} \underbrace{\left[h[\alpha] + 2 \sum_{n=0}^{\alpha-1} h[n] \cos(\omega(\alpha - n)) \right]}_{\text{real, call it } H_r(\omega)}$$

- The bracket is **purely real**, so the entire phase is the exponential in front:

$$|H(e^{j\omega})| = |H_r(\omega)|, \quad \arg H(e^{j\omega}) = -\alpha\omega \quad (\text{plus } \pi \text{ wherever } H_r < 0)$$

- **That is the proof**: the phase is a straight line of slope $-\alpha$, so the group delay is α samples at every frequency.

The antisymmetric case runs identically but the mirror pair subtracts, giving $2jh[n] \sin(\omega(\alpha - n))$ and hence $H = j e^{-j\omega\alpha} H_i(\omega)$. The extra j is the constant $\pi/2$ in the phase, which is why an antisymmetric filter has **generalized** linear phase rather than the plain kind. See §5 of the numerical companion for $h[n] = \{-1, 0, 1\}$, which is this case.

3.7 Last-minute recall

Say these without thinking

- $H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$, **provided the ROC contains the unit circle.**
- $|H| = \frac{\text{product of distances from the zeros}}{\text{product of distances from the poles}}.$
- Pole near the circle $\rightarrow$ peak; zero near the circle $\rightarrow$ dip; zero on the circle $\rightarrow$ exact null.
- Peak sits at the **angle of the pole**, dip at the **angle of the zero.**
- $|H(e^{j0})| = |B(1)/A(1)|$, $|H(e^{j\pi})| = |B(-1)/A(-1)|.$
- Causal and stable $\Leftrightarrow$ **all poles inside** $|z| = 1.$
- FIR: always stable, exactly linear phase possible, all poles at the origin.
- Linear phase $\Leftrightarrow h[n] = \pm h[M - 1 - n]$, delay $\alpha = (M - 1)/2.$

Mistakes that cost marks

- **Forgetting the zeros at the origin** when $M < N.$ They belong on the plot even though they do not shape $|H|.$
- Drawing a magnitude response for a system whose ROC excludes the unit circle. If a pole lies outside and the system is stated causal, **the correct answer is that no frequency response exists.**
- Reading the poles off the x coefficients. **Poles come from the y side.**
- Losing the conjugate. A real difference equation can only have complex roots in **conjugate pairs**; a lone complex pole means an arithmetic slip.
- Sketching a peak at the pole's *radius* instead of its *angle*.
- Quoting ω in Hz. It is in radians per sample, and the plot runs 0 to $\pi.$

4. Discrete Filter Structures

8 hrs · 68 questions across the 45 papers · worth about 10 of 80 marks · 3 syllabus sub-topics · 5th of the 8 chapters by weight, and tied with Ch 1 as the longest at 8 hours.

Read this first:

- 42 of the 68 questions are the **lattice** TOP 42/45, worth 314 of the chapter's 449 marks. It is one calculation, the **step-down recursion**, run on whatever polynomial the paper prints. Learn it once and two thirds of the chapter is finished.
- The other two topics are **realization structures** TOP 16/45 (direct form I and II, cascade, parallel) and **quantization** TOP 10/45 (number formats, round-off, limit cycles).
- **Nothing in this chapter changes what the filter does.** Every structure here realizes the *same* $H(z)$. They differ in how much memory they need, how many multiplies, and how badly finite word length hurts them. That sentence is the answer to "why do we need different structures".
- **The lattice is stable exactly when every $|k_m| < 1$.** That single line answers the "check stability" half of a dozen questions, and it needs no root finding at all.
- A trap worth knowing before you start: a **symmetric** polynomial gives $|k_N| = 1$, and the recursion divides by $1 - k_N^2$. **A linear-phase FIR filter has no lattice realization.** Five papers ask for one anyway; see §4.4.

4.1 Why there is more than one structure

Why do we need different realization structures for the same system? 2 4

the opening paragraph for any structure question

What is the same

- Every structure in this chapter realizes the **same** $H(z)$, so it has the same impulse response, the same poles and zeros, and the same frequency response. With infinite precision they are indistinguishable.
- So a question that asks you to "realize" a system is asking for a **block diagram or signal flow graph**, not for new mathematics.

What differs, and why it matters

- **Memory:** how many delay elements. Direct Form II uses the minimum possible, $\max(M, N)$.
- **Arithmetic:** how many multiplies and adds per output sample.
- **Finite word length:** how far the poles move when the coefficients are rounded to b bits, and how much round-off noise accumulates. **This is the real reason** cascade and parallel forms are preferred in practice: a direct form of high order is extremely sensitive, because every pole depends on every coefficient.
- **Modularity:** cascade and lattice are built from identical sections.

The three building blocks, and the two ways to draw them

- Every structure is made of **unit delays** z^{-1} , **multipliers** (a constant on an arrow) and **adders**.
- A **block diagram** draws adders as circles with $+$; a **signal flow graph** draws nodes and directed branches, with the gain written on the branch. They carry the same information, and papers ask for either.
- **Transposition:** reverse every arrow, swap adders with branch nodes, and exchange input with output. The transposed structure has the *same* $H(z)$. **No paper in the 45 asks for a transposed form**, and no source in the subject covers it, but it is worth one line because it explains how two diagrams that look nothing alike can be the same filter.

4.2 Direct Form I and Direct Form II

Obtain the Direct Form I and Direct Form II realization of the given system 2+2 4 5

TOP 16/45 79 Bh 2+2, 79 Ba 4, 82 Bh 2+2, 74 Ch 5, 72 Ch 5, 72 Ka 4, 69 Ch 4, 68 Bh 9, 70 Ch 4 · worked in §1 of the numerical companion

Direct Form I: the difference equation, drawn

Start from

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k]$$

- Draw the **feedforward** part first: a chain of M delays down the input, each tapped by b_k into a summing node.
- Then the **feedback** part: a chain of N delays down the output, each tapped by $-a_k$ into the same node.
- **It is the equation copied onto paper**, which is why it is the easy 2 marks.
- Cost: $M + N$ delays, $M + N + 1$ multiplies.
- Read it as $H(z) = B(z) \cdot \frac{1}{A(z)}$: all-zero section first, all-pole section second.

Direct Form II: swap the halves and share the delays

Because the two sections are LTI, they commute:

$$H(z) = \frac{1}{A(z)} \cdot B(z)$$

- Put the **all-pole section first**. Its output is an intermediate signal $w[n]$:

$$w[n] = x[n] - \sum_{k=1}^N a_k w[n-k], \quad y[n] = \sum_{k=0}^M b_k w[n-k]$$

- Now **both** chains delay the same signal $w[n]$, so **one chain of delays serves both**.
- Cost: $\max(M, N)$ delays, which is the fewest any structure can use. A structure with the minimum number of delays is called **canonic**.
- **This is the only reason Direct Form II exists**: same arithmetic, half the memory.

Drawing it, in four steps

1. Write the equation with $y[n]$ alone on the left and **watch the signs**: every a_k crosses over and becomes $-a_k$.
2. Count N (highest y delay) and M (highest x delay). The chain has $\max(M, N)$ delays.
3. Down the left of the chain put $-a_1, -a_2, \dots$ feeding back into the top adder. Down the right put $b_0, b_1, \dots$ feeding forward into the output adder.
4. Label $w[n]$ at the top of the chain. A missing coefficient is a **missing branch**, not a zero multiplier: if the paper's equation skips $x[n-1]$, leave that tap out.

If $a_0 \neq 1$, **divide the whole equation by a_0 first**. 74 Ch prints $3y[n] + y[n-1] + 2y[n-4] = 2x[n] + x[n-3]$, and every coefficient must be divided by 3 before drawing. Forgetting this is the most common lost mark in §4.2.

4.3 Cascade and parallel forms

Realize the given $H(z)$ in cascade form of second-order sections, and in parallel form 4 5 6 7

TOP 16/45 81 Ba 5, 74 Ma 7, 80 Ba 4, 76 Ch 4, 77 Ch 7, 69 Bh 6, 67 Mng, 78 Ch 6

Cascade: multiply the sections

$$H(z) = b_0 \prod_{k=1}^K \frac{1 + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}}$$

- Group the poles and zeros into **second-order sections**, each realized in Direct Form II, and chain them output to input.
- **A complex pole must be paired with its conjugate** in the same section. That is the whole point: the section then has **real** coefficients, $a_{1k} = -2\operatorname{Re}(p)$ and $a_{2k} = |p|^2$.

Parallel: add the sections

$$H(z) = C + \sum_{k=1}^K \frac{\gamma_{0k} + \gamma_{1k}z^{-1}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}}$$

- Get here by **partial fractions**, exactly as in Chapter 2, then recombine each conjugate pair into one real second-order section.
- C is the constant from the division when $H(z)$ is improper; it is a direct path from input to output.

- Real poles pair with real poles, or sit alone as a first-order section.
- An odd number of real poles leaves one first-order section; that is normal, not an error.
- **Cascade needs factors, parallel needs partial fractions.** That is the only difference in the work.
- Parallel has the best round-off behaviour of the three: an error in one section cannot propagate into the others, because they do not feed each other.

Turning a conjugate pair into real coefficients

- A pair of poles at $re^{\pm j\theta}$ multiplies out to

$$(1 - re^{j\theta}z^{-1})(1 - re^{-j\theta}z^{-1}) = 1 - 2r\cos\theta z^{-1} + r^2z^{-2}$$

- So a section written $1 - 2r\cos\theta z^{-1} + r^2z^{-2}$ needs only r and θ , both of which the papers usually hand you directly.
- For a pair given in rectangular form $\sigma \pm j\omega$, it is $1 - 2\sigma z^{-1} + (\sigma^2 + \omega^2)z^{-2}$.
- **Check every section has real coefficients before drawing.** A surviving j means a pole was left unpaired.

4.4 The FIR lattice

Compute the lattice coefficients and draw the structure for the given $H(z)$; check stability **5 6 8 10**

TOP 42/45 80 Bh **10**, 81 Ch, 72 Ash **4+2+1**, 79 Ch, 69 Bh **8**, 71 Bh **6+2**, 70 Ma **6**, 71 Ch **4+2+1**, 81 Ba **5**, 78 Bh **3+7**, 79 Bh **6**, 71 Shr **5**, 80 Ch, 74 Ma **7**, and 25 more · **the single most valuable calculation in the subject**

What the structure is

An M -stage lattice chains identical two-input two-output sections. Stage m takes a forward signal f_{m-1} and a backward signal g_{m-1} and produces

$$f_m[n] = f_{m-1}[n] + k_m g_{m-1}[n-1]$$

$$g_m[n] = k_m f_{m-1}[n] + g_{m-1}[n-1]$$

with $f_0[n] = g_0[n] = x[n]$ and $y[n] = f_M[n]$.

- The k_m are the **reflection coefficients**, one per stage.
- The forward path realizes $A_M(z)$; **the backward path realizes its reverse** $z^{-M}A_M(z^{-1})$, which is what makes the all-pass of §4.5 so tidy.
- Only **one** multiplier value per stage, used twice. That symmetry is why the lattice is far less sensitive to coefficient quantization than a direct form, and why it is used for speech.

The step-down recursion (**direct form** → **lattice**)

Given $A_M(z) = 1 + a_M(1)z^{-1} + \dots + a_M(M)z^{-M}$:

$$k_m = a_m(m)$$

$$a_{m-1}(i) = \frac{a_m(i) - k_m a_m(m-i)}{1 - k_m^2}, \quad i = 1, \dots, m-1$$

1. Read off k_M : it is the last coefficient, free.
2. Fold A_M against its own reverse, divide by $1 - k_M^2$, and you have A_{M-1} , one degree lower.
3. Repeat down to A_1 , whose single coefficient is k_1 .

The step-up recursion (**lattice** → **direct form**)

$$a_m(i) = a_{m-1}(i) + k_m a_{m-1}(m-i), \quad a_m(m) = k_m$$

Same relation, rearranged. Papers ask in both directions: 79 Bh and 71 Shr give $k = \{\frac{1}{4}, \frac{1}{2}, \frac{1}{3}\}$ and want the FIR coefficients.

Stability, in one line

- **The system is stable if and only if $|k_m| < 1$ for every m .**
- No root finding, no Jury table. The step-down is the Schur-Cohn test: it is testing whether every root of $A(z)$ lies inside the unit circle.
- A question that says "check stability" after asking for the lattice is asking you to **look at the numbers you already computed**. It is a free mark.

The symmetric trap: why five papers ask for an impossible lattice If $A(z)$ is **symmetric**, meaning its coefficient list reads the same backwards, then $a_M(M) = a_M(0) = 1$, so $k_M = 1$ and the step-down divides by $1 - k_M^2 = 0$. **A symmetric polynomial has no lattice realization.** And symmetry is exactly the linear-phase condition $h[n] = h[M-n]$ from Chapter 3, so a **linear-phase FIR filter can never be put in lattice form**. The papers that hit this are **72 Ch** ($1 + 2z^{-1} + z^{-2}$), **73 Bh** ($1 + 2z^{-1} + 2z^{-2} + z^{-3}$), **75 Ch** and **73 Ch** (both with $1 + \frac{2}{3}z^{-1} + \frac{5}{8}z^{-2} + \frac{2}{3}z^{-3} + z^{-4}$), and **74 Ash**, whose last coefficient is -1 . Say so, show the division by zero, and note that the zeros sit on the unit circle. It is a better answer than a page of arithmetic that cannot finish.

4.5 Lattice-ladder, for a pole-zero system

Why a ladder is needed

- The plain lattice realizes an **all-zero** $A_M(z)$, or upside down an **all-pole** $1/A_N(z)$.
- A general $H(z) = B(z)/A(z)$ needs one more piece: tap every backward signal $g_m[n]$ through a weight C_m and sum them.

$$y[n] = \sum_{m=0}^M C_m g_m[n]$$

- The lattice supplies the poles, **the ladder taps supply the zeros**.

Getting the ladder coefficients

Run the step-down on the **denominator** $A(z)$ and keep every intermediate polynomial $A_i(z)$. Then, working **downwards** from $m = M$:

$$C_m = b_m - \sum_{i=m+1}^M C_i a_i(i-m)$$

1. $C_M = b_M$, because the sum is empty.
2. Each lower C subtracts what the higher taps already contribute.
3. **The k 's come only from $A(z)$** , so two papers with the same denominator have the same lattice and differ only in the ladder. 76 Ch and 76 Ash are such a pair, and so are 74 Bh and 75 Bh.

The all-pass result, and why it is worth remembering 70 Bh gives $H(z) = \frac{1 + 0.4z^{-1} - 1.2z^{-2} + 2z^{-3}}{2 - 1.2z^{-1} + 0.4z^{-2} + z^{-3}}$ and asks how a lattice implements it. Look at the coefficients: **the numerator is the denominator reversed**. That is the definition of an all-pass, $H(z) = z^{-N} A(z^{-1})/A(z)$, and the backward path of a lattice already computes exactly $z^{-N} A(z^{-1})$. So the ladder collapses: every C_m is zero except $C_N = 1$. **An all-pass filter is just a lattice with the output taken from the last backward node**, no ladder taps at all. The reflection coefficients are $k = \{-0.56, \frac{2}{3}, \frac{1}{2}\}$, all less than 1 in magnitude, so it is stable.

4.6 Number representation and quantization

Write about the sign magnitude and 2's complement representation of a binary fractional number. Write about truncation error and rounding error 6 3+4

TOP 10/45 69 Bh 6, 69 Bh 3+4, 75 Ash 3, 67 Mng, 71 Shr 1+1

The three fixed-point codes

A fraction in $[-1, 1)$ with b bits after the binary point. Positive numbers look the same in all three; only negatives differ.

Code	How a negative is formed
Sign magnitude	sign bit 1, magnitude bits unchanged
1's complement	invert every magnitude bit
2's complement	invert every bit, add 1 in the last place

- Worked, for $\pm 5/8$ in 3 bits, $5/8 = 0.101_2$:
 $+5/8 = 0.101$ in all three.
 $-5/8 = 1.101$ (SM), 1.010 (1's), 1.011 (2's).
- **2's complement is used in practice** because addition needs no special case and overflows in a partial sum cancel out.

Rounding against truncation

With step $q = 2^{-b}$ and error $e = Q(x) - x$:

Scheme	Error range
Rounding, any code	$-q/2 \leq e \leq q/2$
Truncation, sign mag.	$-(q - q_u) \leq e \leq (q - q_u)$
Truncation, 2's comp.	$-(q - q_u) \leq e \leq 0$

- **Rounding is better**, and this table is why: its error is centred on zero, so it has **no bias**, and its range is half as wide.
- Truncating a **sign magnitude** number always shrinks the magnitude, so the error is symmetric about zero but takes the sign of x .
- Truncating a **2's complement** number always moves the value **down**, never up, so the error is **one-sided and negative**: a systematic bias, a DC offset that accumulates through a filter.
- $q_u = 2^{-b_u}$ is the step before quantization, with $b_u > b$.

The three places quantization bites

- **Input (A/D) quantization**: modelled as additive white noise of variance $q^2/12$ for rounding.

- **Coefficient quantization:** the a_k and b_k are stored in b bits, so **the poles move**. A high-order direct form is the worst case, because every pole depends on every coefficient; cascade and parallel forms confine the damage to one section, which is the practical argument for §4.3.
- **Product round-off:** each multiply doubles the word length and must be shortened again. In a recursive filter this feeds back, which is §4.7.

4.7 Limit cycles and the dead band

What is the limit cycle effect in a recursive system? Describe it with an example. Define dead band

3 5 1+3

TOP 10/45 71 Ch 3, 70 Asa 1+3, 72 Ma 5, 81 Ch, 80 Ch, 70 Bh 3+4, 66 Ma 4, 71 Shr 1+1+2+1

What a limit cycle is

- In a **recursive** filter the rounded product is fed back and rounded again. Once the signal is small enough, rounding can return the *same magnitude* it started from.
- The filter then never decays: after the input stops it sits in a **self-sustaining oscillation** of fixed amplitude. That is a **zero-input limit cycle**.
- It happens only with quantization. **The ideal infinite-precision filter is perfectly stable**; the limit cycle is created by the arithmetic, not by the poles.
- FIR filters cannot have them: no feedback, nothing to sustain.

The dead band

For the first-order $y[n] = Q\{a y[n-1]\} + x[n]$ with rounding, the filter cannot escape once

$$|y[n]| \leq \frac{q}{2(1-|a|)}$$

- That interval is the **dead band**: inside it, quantization exactly undoes the decay that a should have produced.
- The oscillation amplitude is bounded by the dead band, so a longer word (smaller q) makes it smaller but never removes it.
- If $a > 0$ the trapped value holds **constant**, period 1. If $a < 0$ it **alternates in sign**, period 2.

The worked example, 66 Ma (and the shape of every limit-cycle answer)

$y[n] = Q\{-0.5 y[n-1]\} + x[n]$, rounded to 3 bits so $q = \frac{1}{8}$, with $y[-1] = 0$ and $x[n] = 0.875 \delta[n]$:

n	$-0.5 y[n-1]$	rounded to $\frac{1}{8}$	$+ x[n]$	
0	0	0	0.875	the impulse arrives
1	-0.4375	-0.5	-0.5	
2	0.25	0.25	0.25	
3	-0.125	-0.125	-0.125	now inside the dead band
4	0.0625	0.125	0.125	rounding put it back up
5	-0.0625	-0.125	-0.125	

- From $n = 3$ on it alternates $\pm \frac{1}{8}$ for ever: a limit cycle of amplitude $\frac{1}{8}$ and **period 2**, the period coming from $a < 0$.
- Dead band: $\frac{q}{2(1-|a|)} = \frac{1/8}{2(1-0.5)} = \frac{1}{8}$, which is exactly the amplitude observed. **Quote the formula and the table agrees with it.**
- Step 4 is the whole mechanism in one line: $-0.5 \times (-0.125) = 0.0625$, which is *half* a step, and rounding carries it up to a full 0.125 instead of down to 0.

4.8 Last-minute recall

Say these without thinking

- Direct Form II is **canonic**: $\max(M, N)$ delays, the fewest possible.
- Divide through by a_0 before drawing anything.
- Cascade needs **factors**; parallel needs **partial fractions**.
- A complex pole always sits with its conjugate, giving $1 - 2r \cos \theta z^{-1} + r^2 z^{-2}$.
- Step-down: $k_m = a_m(m)$, then
$$a_{m-1}(i) = \frac{a_m(i) - k_m a_m(m-i)}{1 - k_m^2}.$$
- Step-up: $a_m(i) = a_{m-1}(i) + k_m a_{m-1}(m-i)$.
- **Stable** $\Leftrightarrow$ every $|k_m| < 1$.
- Ladder: $C_m = b_m - \sum_{i>m} C_i a_i(i-m)$, downwards from $C_M = b_M$.
- Dead band = $\frac{q}{2(1-|a|)}$; $a < 0$ gives period 2.

Mistakes that cost marks

- **Forgetting to divide by a_0** when the paper prints $3y[n] + \dots$.
- Sign errors moving a_k across the equals sign: the feedback taps are $-a_k$.
- Running the step-down on the **numerator** of a lattice-ladder question. The k 's come from $A(z)$, always.
- Grinding through a step-down that has already produced $|k| = 1$. Stop and say the polynomial is symmetric.
- Reporting k to two decimals and then calling $|k| = 0.99$ stable without saying so. Keep fractions where the paper gives fractions.
- Claiming an FIR filter can have a limit cycle. It cannot: no feedback.
- Saying truncation error is symmetric for 2's complement. **It is one-sided**, and that is the point of the question.

5. FIR Filter Design

6 hrs · 70 questions across the 45 papers · worth about 12 of 80 marks · 3 syllabus sub-topics · 4th of the 8 chapters by weight.

Read this first:

- Three topics, and they divide cleanly by *what you have to do*:
 - Window method** TOP 28/45: a design calculation. Pick a window, get a length, truncate the ideal response.
 - Kaiser window** TOP 19/45: the same calculation with two formulas bolted on, for β and for the length.
 - Remez exchange** TOP 23/45: **pure bookwork**. 23 questions, almost always the same flowchart and the same three sentences, and **not one of them asks you to run the algorithm**.
- So a third of this chapter's marks are free if you memorise one flowchart. Do §5.6 first.
- The whole design method is two lines**: the ideal filter has an infinite, non-causal impulse response, so **truncate it** and **taper the truncation** so the edges do not ring.
- The window is chosen by **stopband attenuation** and the length by **transition width**. Those are two independent lookups and they are the only two numbers a fixed-window question turns on.
- Gibbs ripple does not shrink when you lengthen the filter**. That single fact is worth 3 to 6 marks in eight papers, and it is the reason Kaiser exists.

5.1 The window method

Describe how a digital FIR filter is designed by the window method, and discuss the characteristics of the different window functions 4+4 5+3 4 5

TOP 28/45 70 Ch 4+4, 72 Ch 5+3, 72 Ma 4, 70 Bh 5 · the answer is the four steps below, then the table in §5.2

Where the ideal response comes from

An ideal low-pass filter is a brick wall in frequency. Inverse DTFT it:

$$h_d[n] = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{\sin \omega_c n}{\pi n}$$

- This is a sinc: **infinitely long** and **non-causal**, so it cannot be built.
- Delay it by $\alpha = \frac{N-1}{2}$ samples to make it causal, which is also exactly what gives **linear phase**:

$$h_d[n] = \frac{\sin(\omega_c(n-\alpha))}{\pi(n-\alpha)}, \quad h_d[\alpha] = \frac{\omega_c}{\pi}$$

- The centre tap is a 0/0 limit, so **evaluate it separately as ω_c/π** . Forgetting that is the most common slip in the chapter.

Why multiplying by a window is convolving in frequency

- $h[n] = h_d[n]w[n]$ becomes, in frequency,

$$H(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta$$

- So the brick wall is **smeared by the window's spectrum**. The result is the sharp edge replaced by a transition, plus ripple from the window's sidelobes.
- Main lobe width sets the transition; sidelobe height sets the ripple**. That one sentence explains the entire comparison table in §5.2.

The four steps, which are the whole method

- Write $h_d[n]$ for the ideal filter, delayed by $\alpha = (N-1)/2$.
- Choose the window** from the required stopband attenuation.
- Choose N** from the required transition width.
- Multiply**: $h[n] = h_d[n]w[n]$ for $0 \leq n \leq N-1$, zero elsewhere.

Other ideal responses

- High pass**: $\delta[n-\alpha] - h_{d,LP}[n]$, an all-pass minus the low pass.
- Band pass**: $h_d(\omega_2) - h_d(\omega_1)$, the wider low pass minus the narrower.
- Every one of them is symmetric about α** , so every windowed FIR filter here has linear phase automatically.

5.2 The windows, and the Gibbs phenomenon

Explain the Gibbs phenomenon: how it arises when the rectangular window is used, and how it can be minimized 3+3 5 6 1+4 2+5

TOP 8/45 81 Ba, 80 Ba, 73 Shr, 71 Shr, 79 Ch, 74 Bh, 71 Bh, 81 Ch · asked in eight papers, always the same answer

What the Gibbs phenomenon is

- Truncating the sinc with a **rectangular** window convolves H_d with the Dirichlet kernel, whose sidelobes are only **13 dB** below its main lobe.
- The result overshoots the band edge by about **9 % of the discontinuity height**, with ringing either side.
- **Increasing N does not help.** The ripples get narrower and more numerous, but the overshoot stays at 9 %. The series converges in the mean-square sense, **not uniformly**.
- That is the whole phenomenon, and the fact examiners want stated.

How to minimize it

- **Use a tapered window.** Hann, Hamming and Blackman fall smoothly to zero at the ends instead of being cut off, so their sidelobes are far lower and the ripple is far smaller.
- The price is a **wider main lobe**, hence a wider transition band, hence a longer filter for the same sharpness.
- **Every window is a trade of ripple against transition width.**

The window formulas, for $0 \leq n \leq M$ with $M = N - 1$

$$w_{\text{rect}}[n] = 1, \quad w_{\text{Bart}}[n] = 1 - \frac{|2n - M|}{M}$$

$$w_{\text{Hann}}[n] = 0.5 - 0.5 \cos \frac{2\pi n}{M}, \quad w_{\text{Hamm}}[n] = 0.54 - 0.46 \cos \frac{2\pi n}{M}$$

$$w_{\text{Black}}[n] = 0.42 - 0.5 \cos \frac{2\pi n}{M} + 0.08 \cos \frac{4\pi n}{M}$$

- **Hann ends at exactly 0; Hamming ends at 0.08.** That is the only difference between them and it buys 9 dB of stopband.
- All five are **symmetric**, which is what keeps the phase linear.

The comparison table (learn this: it answers half the chapter)

Window	Main lobe	Peak side lobe	Stop band	Transition
Rectangular	$\frac{4\pi}{M+1}$	−13 dB	21 dB	$1.8\pi/M$
Bartlett	$8\pi/M$	−25 dB	25 dB	$6.1\pi/M$
Hann	$8\pi/M$	−31 dB	44 dB	$6.2\pi/M$
Hamming	$8\pi/M$	−41 dB	53 dB	$6.6\pi/M$
Blackman	$12\pi/M$	−57 dB	74 dB	$11\pi/M$

- **Stopband attenuation is a property of the window alone.** It does not depend on N at all, which is exactly why Gibbs ripple cannot be lengthened away.
- $M = N - 1$ throughout. Papers sometimes call M the length; read which.
- The two columns you actually use are **Stop band** (to choose the window) and **Transition** (to choose N).

5.3 Choosing the window and the length

The two lookups

- Window** ← stopband attenuation. Take the *cheapest* window whose figure meets the spec. If the paper gives a stopband ripple δ_s instead, convert first:

$$A_s = -20 \log_{10} \delta_s \text{ dB}$$

- Length** ← transition width $\Delta\omega = |\omega_s - \omega_p|$, from the same table:

$$N - 1 = M = \frac{c \pi}{\Delta\omega}$$

with c the window's transition constant (6.6 for Hamming, and so on).

- Round N **up**, and up again to the next **odd** number so $\alpha = (N - 1)/2$ is a whole sample.

The cut-off goes in the middle

- The window method has no independent control of the band edges, so put the ideal cut-off **halfway across the transition band**:

$$\omega_c = \frac{\omega_p + \omega_s}{2}$$

- Use ω_c , not ω_p , in $h_d[n]$.** Using the passband edge is a guaranteed lost mark.
- If the spec is in Hz, convert first: $\omega = 2\pi f/f_s$. If it is in rad/s, $\omega = \Omega/f_s$ (equivalently ΩT).

The window choice is a step, not a slope. 67 Mng asks for 54 dB.

Hamming gives 53, one short, so Blackman is forced and the filter jumps from 67 taps to 111 for the sake of 1 dB. Say so: it is the point of the question.

5.4 The Kaiser window

Why is the Kaiser window better than other fixed windows? Explain the steps to design an FIR filter using it 3 4 7 2+2

TOP 19/45 81 Ba, 74 Ash, 72 Ch, 80 Ch, 72 Ash, 74 Bh 2+2 · the "why" is one sentence, the design is four lines

Why it is better (the whole answer)

- Every fixed window offers **one** stopband attenuation, take it or leave it. Hann gives 44 dB whether you want 30 or 43.
- Kaiser has a **shape parameter** β , so the attenuation is *continuously adjustable*.
- So the two specifications **decouple**: β **buys attenuation**, N **buys transition width**, and neither is wasted on the other.
- The result is **the shortest filter** that meets a given spec among all the window methods, because nothing is over-designed.
- It is also very nearly optimal in the energy sense: it maximises the fraction of the window's energy in the main lobe.

The design equations

$$w[n] = \frac{I_0\left(\beta \sqrt{1 - \left(\frac{2n}{M} - 1\right)^2}\right)}{I_0(\beta)}, \quad 0 \leq n \leq M$$

- $\delta = \min(\delta_p, \delta_s)$, then $A = -20 \log_{10} \delta$ dB.

$$\beta = \begin{cases} 0 & A \leq 21 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21) & 21 < A \leq 50 \\ 0.1102(A - 8.7) & A > 50 \end{cases}$$

$$M = \frac{A - 8}{2.285 \Delta\omega}, \quad \text{length} = M + 1$$

- I_0 is the **modified Bessel function of the first kind**, order

zero: $I_0(x) = \sum_{k \geq 0} \left[\frac{(x/2)^k}{k!} \right]^2$, and $I_0(0) = 1$.

$\beta = 0$ makes the Kaiser window rectangular, and larger β tapers it harder.

The same δ gives the same β , so most papers in §2 of the companion share one value.

5.5 Optimum filters and the Remez exchange algorithm

What is an optimum filter? Explain the Remez exchange algorithm for FIR filter design, with its flowchart. Why is it superior to windowing? 1+5 2+6 4 9 3+10

TOP 23/45 75 Ch, 72 Ch, 70 Ch, 69 Ch, 73 Ch, 78 Bh, 76 Ch, 82 Bh, 81 Bh, 76 Ash, 73 Shr, 71 Shr, 79 Bh, 79 Ba, 82 Ba, 77 Ch, 74 Bh, 78 Ch, 80 Ch, 74 Ma, 72 Ma, 70 Bh, 69 Bh · **23 questions, 175 marks, and no calculation in any of them**

What "optimum" means

- The window method accepts whatever error the window happens to give, and that error is **largest at the band edges** and small in the middle. Most of the allowed tolerance goes unused.
- An **optimum** (equiripple, minimax, Parks-McClellan) filter instead spreads the error evenly, so the worst case is as small as possible:

$$\min_h \max_{\omega \in S} |E(\omega)|, \quad E(\omega) = W(\omega) [H_d(\omega) - H(\omega)]$$

- $W(\omega)$ is a **weight**, which is how passband and stopband ripple can be traded: set $W = \delta_s/\delta_p$ in the passband and 1 in the stopband.
- S is the bands of interest only, so **the transition band is simply left out** rather than being forced to behave.

The alternation theorem, which is why it works

- H is the unique best approximation **if and only if** $E(\omega)$ has at least $r + 2$ **alternations**: extrema of equal magnitude and alternating sign, where r is the number of free coefficients.
- So the answer is recognisable when you see it: **count the alternations**.
- That is what makes the filter *equiripple*: every ripple in a band reaches exactly the same height.

Why it beats windowing

- **Shorter filter** for the same specification. **The local worked comparison is worth quoting**: for $\omega_p = 0.4\pi$, $\omega_s = 0.6\pi$ and $\delta = 0.0116$, the equiripple design needs $M = 26$ while a Kaiser window at the same δ needs $M = 38$. **A third fewer taps**.
- **Independent control** of passband and stopband ripple through W .
- Handles arbitrary multi-band shapes, not just the standard four.

The Remez exchange algorithm, as a flowchart (this is the marks)

- 1. Input** the band edges, the desired response H_d , the weights W , and the order.
- 2. Guess** an initial set of $r + 2$ extremal frequencies, usually spaced evenly over the bands.
- 3. Solve** for the deviation δ on that set. It comes out in closed form, with no matrix inversion:

$$\delta = \frac{\sum_k \gamma_k H_d(\omega_k)}{\sum_k (-1)^k \gamma_k / W(\omega_k)}$$

- 4. Interpolate** the trial response H through the points using Lagrange interpolation, and evaluate $E(\omega)$ on a dense grid.
- 5. Exchange**: find the new local extrema of $|E|$ and make them the new extremal set.
- 6. Test**: if the largest $|E|$ on the grid no longer exceeds $|\delta|$, the alternation theorem is satisfied, so **stop**. Otherwise go back to step 3.
- 7. Output** the impulse response from the converged frequency response.

Each pass can only increase $|\delta|$ towards its optimum, so the iteration always converges. **Draw it as a loop with the test at the bottom**, label the exchange step, and the flowchart marks are safe. No paper in the 45 asks for a numerical iteration.

5.6 Last-minute recall

Say these without thinking

- $h_d[n] = \frac{\sin(\omega_c(n - \alpha))}{\pi(n - \alpha)}$, centre tap ω_c/π , $\alpha = \frac{N-1}{2}$.
- $\omega_c = \frac{\omega_p + \omega_s}{2}$, and $\Delta\omega = |\omega_s - \omega_p|$.
- Window from **attenuation**: 21 / 25 / 44 / 53 / 74 dB.
- Length from **transition**: 1.8 / 6.1 / 6.2 / 6.6 / 11, all $\times \pi/\Delta\omega$.
- $A = -20 \log_{10} \delta$. Kaiser:
 $\beta = 0.5842(A - 21)^{0.4} + 0.07886(A - 21)$ for
 $21 < A \leq 50$, and $M = \frac{A - 8}{2.285\Delta\omega}$.
- Gibbs overshoot is **9 %** and **does not depend on N** .
- Alternation theorem: $r + 2$ equal-magnitude alternating extrema.
- $\omega = 2\pi f/f_s$ for a spec in Hz.

Mistakes that cost marks

- Using ω_p instead of ω_c in $h_d[n]$.
- **Evaluating the centre tap as 0/0** instead of ω_c/π .
- Choosing the window from the transition width and the length from the attenuation. It is the other way round.
- Taking an even N , which puts α half way between samples.
- Saying Gibbs ripple can be fixed by a longer filter. **It cannot.**
- Quoting δ_p when the spec gives δ_s smaller: Kaiser uses $\min(\delta_p, \delta_s)$.
- Writing J_0 for the Bessel function in the Kaiser window. **It is I_0** , the *modified* one, and one paper mislabels it.

6. IIR Filter Design

6 hrs · 50 questions across the 45 papers · worth about 13 of 80 marks · 4 syllabus sub-topics · 2nd of the 8 chapters by weight, behind only the DFT.

Read this first:

- **36 of the 50 questions are one calculation:** design an analog Butterworth low-pass filter, then map it to digital. Learn that one procedure and the chapter is mostly done.
- The whole subject of IIR design is **two sentences**. There is no good direct method in the z domain, but analog filter design is a solved, closed-form art. So **design in s , then map $s \rightarrow z$** .
- **Only two maps exist** in this syllabus, and they split the chapter:
 - **Bilinear** TOP 36/45: $s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$. One to one, so **no aliasing**, but the frequency axis is **warped**, so the edges must be pre-warped first.
 - **Impulse invariance** TOP 11/45: $h[n] = h_a(nT)$. The frequency axis is **linear**, so no warping, but sampling an unbandlimited analog response **aliases**.
- **Everything else is the analog half:** which prototype (Butterworth or Chebyshev), what order, what cut-off. That part is identical for both maps.
- **The sampling period cancels.** When the spec is given in rad/sample, pre-warp by $\Omega = \frac{2}{T} \tan \frac{\omega}{2}$ and map back by $s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$, and **every $2/T$ divides out:** $H(z)$ is the same for $T = 1$, $T = 2$ or $T = 0.1$. Nine papers give a sampling frequency purely as a distraction. **Set $T = 1$ and write one line saying why** (§6.5).
- The other 14 questions are bookwork: why transform at all, the two methods compared, pre-warping, and spectral transformation. All short, all below.

6.1 Why design through an analog prototype

Why do we use transformation of continuous time filter design methods to design discrete time IIR filters? 3+6 3

PIN 2/45 70 Ma 3+6, 73 Shr 3 · three reasons, one line each, then move on

The three reasons

- 1. The analog art is mature.** Continuous-time IIR design is a century old and its results are tabulated. Reusing it is free.
- 2. Closed-form formulas exist.** Butterworth, Chebyshev and elliptic prototypes all have closed-form order, cut-off and pole expressions, so the design is arithmetic, not search.
- 3. No such formulas exist in the z domain.** The same approximation methods applied directly to a discrete-time filter do *not* give closed forms, because a digital frequency response is **periodic** and an analog one is not.

The three steps that follow from it

- 1. Translate the spec** from digital ω to analog Ω , by the rule that belongs to the map being used.
- 2. Design $H_a(s)$:** choose the prototype family, find N and Ω_c , place the poles, write $H_a(s)$.
- 3. Map back** to $H(z)$ with the same rule.

What the map must preserve

- The $j\Omega$ axis of the s plane must land on the **unit circle** of the z plane, or the frequency response is not preserved.
- The **left half plane** must land **inside** the unit circle, or a stable analog filter becomes an unstable digital one.
- **Both maps below satisfy both conditions.** That is the whole reason there are only two of them.

6.2 The analog prototypes

Compare the Butterworth approximation with the Chebyshev approximation 1+10 3

PIN 2/45 76 Bh 1+10, 82 Bh · a table, not prose

Butterworth: maximally flat

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

- **All pole.** Every zero is at $\Omega \rightarrow \infty$.
- $|H_a(j0)| = 1$ for every N , and $|H_a(j\Omega_c)| = \frac{1}{\sqrt{2}}$, i.e. **Ω_c is the 3 dB point whatever the order.**
- **Monotonic** in both bands: no ripple anywhere.
- Poles lie on a **circle** of radius Ω_c , equally spaced, none on the $j\Omega$ axis.

Chebyshev type I: equiripple passband

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \varepsilon^2 V_N^2(\Omega/\Omega_c)}$$

- $V_N(x) = \cos(N \cos^{-1} x)$ for $|x| \leq 1$, which oscillates between ± 1 , and $\cosh(N \cosh^{-1} x)$ for $x > 1$, which climbs fast.
- So the passband **ripples** between 1 and $1/\sqrt{1 + \varepsilon^2}$, and the stopband falls monotonically and **steeply**.
- **Type II** puts the ripple in the stopband instead, by using V_N^{-1} ; **elliptic** ripples in both and is steeper still.
- $|H_a(j0)| = 1$ **for N odd but $1/\sqrt{1 + \varepsilon^2}$ for N even.** An even-order Chebyshev filter does **not** have unity DC gain.

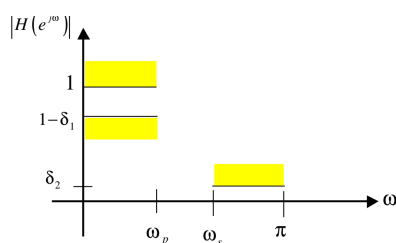
The comparison, as the answer

	Butterworth	Chebyshev I
Passband	monotonic, maximally flat	equiripple between 1 and $1/\sqrt{1 + \varepsilon^2}$
Stopband	monotonic	monotonic
Transition	widest of the three families	narrower for the same order
Order for a given spec	highest	lower, typically by a third
Poles lie on	a circle, radius Ω_c	an ellipse , semi-axes $a\Omega_c$ and $b\Omega_c$
Phase	most nearly linear of the three	worse, and worst near the band edge
Use when	flatness matters more than sharpness	a short filter matters more than flatness

The number to quote comes from the 45 papers' own recurring spec ($\omega_p = 0.2\pi$, $\omega_s = 0.3\pi$, 1 dB, 15 dB), asked by 73 Shr, 77 Ch and 76 Bh: **Butterworth needs $N = 6$, Chebyshev only $N = 4$.** That single sentence answers "compare" better than a paragraph. verified/added The local notes offer a second example, $\Omega_p = 1$, $\Omega_s = 3.33$, $\alpha_p = 0.3$ dB, $\alpha_s = 22$ dB, and print " $n = 5$ for butterworth, $n = 3$ for chebyshev". **The Butterworth figure is wrong:** the ratio is $157.489/0.071519 = 2202.05$, its log is 3.3429, and $2 \log(3.33) = 1.0449$, so $N = 3.1992$ and the answer is **4, not 5**. The Chebyshev 3 is right. Use the paper-derived example above and this one only with the correction.

The design formulas, in the form every paper's specification uses core

these four lines carry 36 questions · memorise them exactly



Digital filter specifications

The tolerance scheme that every paper states in words. δ_1 is the passband deviation, δ_2 the stopband gain, and **only the magnitude is ever specified**: an IIR filter's phase is whatever the poles give.

Attenuation is always positive dB

$$\alpha = -20 \log_{10} |H|$$

- "maximum deviation 1 dB below 0 dB gain" $\Rightarrow \alpha_p = 1$.
- "maximum gain -15 dB in the stopband" $\Rightarrow \alpha_s = 15$.
- a gain 0.89125 given directly $\Rightarrow \alpha_p = -20 \log_{10} 0.89125 = 1$ dB.
- a ripple $\delta_p = \delta_1 = 0.11 \Rightarrow$ the edge gain is $1 - \delta_p = 0.89$, so $\alpha_p = 1.0122$ dB; but $\delta_s = \delta_2 = 0.21$ is a gain already, so $\alpha_s = 13.5556$ dB.
- **Those four phrasings are the same spec four ways.** Converting first, before anything else, is what makes the rest mechanical.

Order and cut-off

$$N \geq \frac{\log_{10} \left[\frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1} \right]}{2 \log_{10}(\Omega_s/\Omega_p)} \quad (\text{Butterworth})$$

$$N \geq \frac{\cosh^{-1} \sqrt{\frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1}}}{\cosh^{-1}(\Omega_s/\Omega_p)} \quad (\text{Chebyshev})$$

$$\Omega_c = \frac{\Omega_p}{(10^{0.1\alpha_p} - 1)^{1/2N}}, \quad \varepsilon = \sqrt{10^{0.1\alpha_p} - 1}$$

- **Always round N UP.** The fractional N meets the spec exactly and is not realisable.
- That Ω_c makes the **passband** exact and leaves the stopband over-satisfied. Using Ω_s and α_s instead makes the stopband exact. Both are legal; **state which you used.**
- Chebyshev needs no separate Ω_c : the passband is equiripple right up to Ω_p , so $\Omega_c = \Omega_p$.

Chebyshev poles

$$\alpha = \frac{1}{\varepsilon} + \sqrt{1 + \frac{1}{\varepsilon^2}}, \quad a = \frac{\alpha^{1/N} - \alpha^{-1/N}}{2}, \quad b = \frac{\alpha^{1/N} + \alpha^{-1/N}}{2}$$

$$s_k = -a\Omega_c \sin \theta_k + j b\Omega_c \cos \theta_k, \quad \theta_k = \frac{(2k+1)\pi}{2N}$$

- The poles sit on the **ellipse** $\frac{\sigma^2}{a^2\Omega_c^2} + \frac{\Omega^2}{b^2\Omega_c^2} = 1$, at the same angles a Butterworth filter would use, squashed towards the $j\Omega$ axis.
- $b > 1 > a$ always, so the ellipse is **taller than wide**: the poles crowd the imaginary axis, which is what sharpens the cut-off.

Butterworth poles

$$s_k = \Omega_c e^{j\pi \frac{N+2k+1}{2N}}, \quad k = 0, 1, \dots, N-1$$

- These are the N left-half-plane roots of $1 + (s/j\Omega_c)^{2N} = 0$; the other N belong to $H_a(-s)$ and are discarded.
- Equally spaced by π/N on the circle of radius Ω_c , symmetric about the real axis, **never on the $j\Omega$ axis.**
- A real pole at $-\Omega_c$ appears **only for odd N .**

$$H_a(s) = \frac{\Omega_c^N}{\prod_k (s - s_k)}$$

- The numerator Ω_c^N is fixed by $H_a(0) = 1$, not chosen.

Chebyshev gain, and the even-order trap

- $H_a(s) = \frac{K}{\prod_k (s - s_k)}$ with $K = \prod_k (-s_k)$ for **odd N** , which makes $H_a(0) = 1$.
- For **even N** the equiripple passband starts at its *lower* edge, so $H_a(0) = \frac{1}{\sqrt{1 + \varepsilon^2}}$ and **K is divided by $\sqrt{1 + \varepsilon^2}$.**
- Worked instance: 72 Ma has $\varepsilon = 1$, so its $N = 2$ filter has a DC gain of $1/\sqrt{2} = 0.707$, **not 1**. It looks like an arithmetic error and is not.
- ε and Ω_c come entirely from the passband. **Only N uses the stopband at all.**

6.3 The impulse invariance method

Describe digital Butterworth filter design using the impulse invariance technique. What are the limitations of the impulse invariance technique? **15 12+3**

TOP 11/45 69 Bh **15**, 78 Bh **12+3**, 72 Ch **3+12**, 81 Ch, 80 Ch, 69 Bh **12**, 70 Asa **13**, 74 Ch **12**, 67 Mng **15** · the limitations are half the marks

In one line: **make the digital impulse response the sampled analog one**, $h[n] = h_a(nT)$. Everything below follows from that single sentence, and so does everything wrong with the method.

The idea, and the whole derivation

- **Sample the analog impulse response:** $h[n] = h_a(nT)$. Nothing else is assumed.
- Expand $H_a(s)$ in partial fractions over its N simple poles p_k :

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - p_k} \Rightarrow h_a(t) = \sum_k A_k e^{p_k t} u(t)$$

- Sample it and take the z transform, which is just a geometric series in $e^{p_k T} z^{-1}$:

$$H(z) = \sum_{n=0}^{\infty} \left(\sum_k A_k e^{p_k nT} \right) z^{-n} = \boxed{\sum_{k=1}^N \frac{A_k}{1 - e^{p_k T} z^{-1}}}$$

- So the rule is **term by term**: $\frac{1}{s - p_k} \rightarrow \frac{1}{1 - e^{p_k T} z^{-1}}$. That is the only thing to remember.
- **Spec translation is linear**: $\Omega = \omega/T$. **No pre-warping**.
- **The T convention**. Some texts set $h[n] = T h_a(nT)$ instead, which multiplies every A_k by T . Without it the filter's gain comes out near $1/T$, so with $T \neq 1$ you must say which you used. **With $T = 1$ s the two are identical**, and 8 of these 11 papers set $T = 1$.

The limitations (say all four)

1. **Aliasing**. Sampling gives $H(e^{j\omega}) = \frac{1}{T} \sum_k H_a(j\frac{\omega+2\pi k}{T})$. No practical analog prototype is band-limited, so the tails fold back and the response is **never exactly the analog one**.
2. **Low pass and band pass only**. A high-pass or band-stop prototype has an infinite high-frequency tail, so it aliases catastrophically. **Impulse invariance cannot make a high-pass filter**.
3. **Needs a partial fraction expansion** of $H_a(s)$, and it assumes the poles are **simple**. Repeated poles need a separate rule.
4. **Gain depends on T** , so the result must be rescaled.

What the map does to the planes

- Each pole moves as $p_k \rightarrow e^{p_k T}$, so $|z| = e^{\sigma_k T}$ and $\angle z = \Omega_k T$.
- $\sigma_k < 0 \Rightarrow |z| < 1$: **a stable analog filter always gives a stable digital one**.
- But $z = e^{sT}$ is **many to one**: every horizontal strip of width $2\pi/T$ in the s plane maps onto the whole z plane.
- **This applies to the POLES only, not to H** . $H(z) \neq H_a(s)|_s$ under any substitution; only the pole positions transform that simply.

The spec translation, which is all of step 1

- Take the tolerance scheme of §6.2 and **divide every frequency edge by T** . Nothing else changes: the bands $1 - \delta_1$ and δ_2 are untouched.
- The prototype is then specified only up to π/T , above which it is *assumed* negligible. **That assumption is exactly what aliasing violates**, which is why the design is checked at the end rather than trusted.
- Contrast §6.4, where the axis is **bent** rather than merely rescaled.
- Practical consequence: **over-design the stopband**. Extra attenuation there is what buys back the margin that the folded tails eat.

6.4 The bilinear transformation

Explain briefly the bilinear transformation method of IIR filter design, with its advantages and disadvantages **3 3+6 2+10**

TOP 36/45 73 Shr **3**, 70 Ma **3+6**, 81 Ba **2+10**, and the 32 design papers in §6.5

Where the formula comes from

- Take the simplest analog filter, $H_a(s) = \frac{b}{s+a}$, i.e.

$$\frac{dy}{dt} + ay(t) = bx(t).$$

- Integrate over one sample interval and apply the **trapezoidal rule**:

$$y(nT) - y(nT - T) = \frac{T}{2} [g(nT) + g(nT - T)], \quad g = bx - ay$$

- Take z transforms and solve. The result is $H_a(s)$ with

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

- So **the bilinear transformation IS trapezoidal integration**, which is why it is stable and one to one. That sentence is worth stating: it is the only derivation in the chapter and papers ask for it.
- Inverting, $z = \frac{1 + sT/2}{1 - sT/2}$.

What it does to the planes

- Put $s = \sigma + j\Omega$. Then

$$|z|^2 = \frac{(1 + \sigma T/2)^2 + (\Omega T/2)^2}{(1 - \sigma T/2)^2 + (\Omega T/2)^2}$$

- $\sigma < 0 \Rightarrow |z| < 1$, $\sigma = 0 \Rightarrow |z| = 1$, $\sigma > 0 \Rightarrow |z| > 1$.

The entire left half plane maps into the unit disc, exactly once.

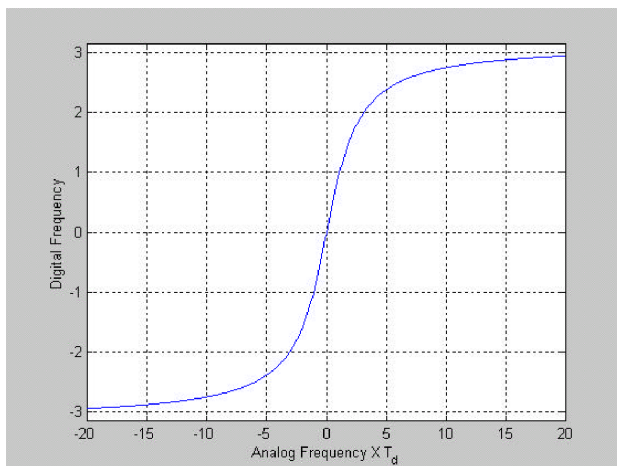
- So stability is preserved, and **there is no aliasing**: the map is one to one, not many to one.
- On the axis, $s = j\Omega$ gives $z = e^{j\omega}$ with

$$\omega = 2 \tan^{-1} \frac{\Omega T}{2},$$

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$$

What is frequency warping, what is pre-warping, and why is it necessary **3+12 3**

HOT 4/45 78 Bh **3**, 82 Bh **3+12**, 72 Ch **3**, 79 Ba **3** · the figure below is the answer



Digital ω against analog ΩT_d . The curve is $\omega = 2 \tan^{-1}(\Omega T_d/2)$: a straight line through the origin at low frequency, flattening to the horizontal asymptotes $\pm\pi$.

Warping

- Ω runs over all of $(-\infty, \infty)$ but ω only over $(-\pi, \pi)$, so **the infinite analog axis is compressed into a finite digital one**. That compression is the warping.
- It is **almost linear for small ω** ($\tan x \approx x$) and **severely non-linear near $\omega = \pi$** .
- Consequence: a shape defined by more than its band edges, such as a differentiator or an arbitrary magnitude curve, comes out **distorted**. Bilinear design is only safe for **piecewise-constant** magnitude responses: low pass, high pass, band pass, band stop.

Pre-warping

- Because the map bends the axis, designing the analog filter at the *stated* edges puts the digital edges somewhere else.
- So **push the edges through the same bend first**: $\Omega_p = \frac{2}{T} \tan \frac{\omega_p}{2}$, $\Omega_s = \frac{2}{T} \tan \frac{\omega_s}{2}$.
- The warp then undoes the pre-warp exactly, and **the critical frequencies land where the specification asked**. Everything between them is still bent, which is why only the edges can be guaranteed.
- Skipping the pre-warp is the single most common lost mark in the chapter.**

6.5 Impulse invariance against bilinear

Compare the impulse invariance method with the bilinear transformation method 12+3 11+3 3+12 2+10

HOT 6/45 81 Bh 12+3, 79 Ba 11+3, 82 Ba 12+3, 72 Ch 3+12, 81 Ba 2+10, 70 Ma 3+6 · asked as a rider on a design question six times

	Impulse invariance	Bilinear
Basis	$h[n] = h_a(nT)$, a time-domain match	trapezoidal integration, a frequency-domain map
$s \rightarrow z$	poles only: $p_k \rightarrow e^{p_k T}$	$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$ throughout
Frequency map	$\Omega = \omega/T$, linear	$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$, non-linear
Mapping type	many to one	one to one
Aliasing	yes , and unavoidable	none
Warping	none, shape preserved	yes, needs pre-warping
Filter types	low pass, band pass only	all four
Non-constant shapes	can do a band-limited differentiator	cannot
Needs	partial fractions, simple poles	algebraic substitution only
Order needed	higher, to absorb the aliasing	lower

The one-sentence verdict – Bilinear trades shape fidelity for the absence of aliasing; impulse invariance does the reverse. – Since almost every filter wanted is piecewise constant, where shape fidelity between the edges is worth nothing, bilinear wins almost always, and 36 of this chapter's 50 questions use it. – The exception is a filter whose <i>shape</i> matters across the band. Only impulse invariance can make a band-limited differentiator.	Why T cancels in a bilinear design verified/added – Pre-warping gives $\Omega_p = \frac{2}{T} \tan \frac{\omega_p}{2}$, so Ω_s/Ω_p is free of T and so is N. – Ω_c then carries one factor $\frac{2}{T}$, so every pole does, so $H_a(s)$ is a function of $sT/2$ alone. – Substituting $s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$ cancels it. $H(z)$ is identical for every T. – So a paper's "sampling frequency 0.5 Hz" changes nothing in the answer. Use $T = 1$, and say in one line that it cancels to show it was not ignored. – This is false for impulse invariance, where $\Omega = \omega/T$ scales the poles but $e^{p_k T}$ does not cancel it.
--	--

6.6 Spectral transformation in the digital domain

Why is spectral transformation required, and what are its features and parameters for low pass to high pass conversion 2 4 11+4

HOT 6/45 70 Asa 2, 80 Ba 4, 70 Ch 5, 79 Bh, 71 Ch, 69 Ch 11+4

Why it is required

- Every design procedure above produces a **low pass** filter. High pass, band pass and band stop have to come from somewhere.
- The analog route (transform $H_a(s)$ into a high pass, then discretise) works for bilinear but **fails for impulse invariance**, because a high-pass analog filter is not band-limited and aliases.
- So do it the other way: **discretise the low pass first**, then transform **inside the z domain**. Aliasing is then impossible, because nothing is sampled.
- **For bilinear designs the two routes give the identical filter**; for impulse invariance only the digital route exists. That is the answer to "why required".

What the map must satisfy

- $p^{-1} = G(z^{-1})$ must be **rational** in z^{-1} .
- It must take the unit circle to the unit circle, and the inside to the inside, so that **stability survives**.
- The general form that does both is an **all-pass** function, $\prod_k \frac{z^{-1} - \alpha_k}{1 - \alpha_k^* z^{-1}}$ with every $|\alpha_k| < 1$.
- **N all-pass factors give an N -fold map**, which is why the two band cases below need z^{-2} and the two low-pass-like ones need only z^{-1} .

The four transformations (this is the table they want)

To	Substitute $z^{-1} \rightarrow$	Parameters
Low pass	$\frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$	$\alpha = \frac{\sin \frac{\omega_p - \omega'_p}{2}}{\sin \frac{\omega_p + \omega'_p}{2}}$
High pass	$-\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}$	$\alpha = -\frac{\cos \frac{\omega_p + \omega'_p}{2}}{\cos \frac{\omega_p - \omega'_p}{2}}$
Band pass	$-\frac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	$a_1 = \frac{2\alpha k}{k+1},$ $a_2 = \frac{k-1}{k+1}$
Band stop	$\frac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	$a_1 = \frac{2\alpha}{k+1},$ $a_2 = \frac{1-k}{1+k}$

For the two band cases $\alpha = \frac{\cos \frac{\omega_u + \omega_L}{2}}{\cos \frac{\omega_u - \omega_L}{2}}$, and $k = \cot \frac{\omega_u - \omega_L}{2} \tan \frac{\omega_p}{2}$ for band pass, $\tan \frac{\omega_u - \omega_L}{2} \tan \frac{\omega_p}{2}$ for band stop. ω_p is the prototype's edge, ω'_p the new one.

Reading the table without mistakes

- **Only the low-pass row keeps the sign**; the high-pass row carries a leading minus, and that minus is the whole reason the passband moves to $\omega = \pi$.
- The substitution goes into $H_{lp}(p)$ **written in p^{-1}** , not in p . Every published prototype already is.
- **Every zero at $z = -1$ becomes a zero at $z = +1$** , so a high pass built this way has $H(1) = 0$ exactly: it kills DC. **Free check on the answer.**
- $|\alpha| < 1$ always. If it comes out bigger, the two edge frequencies have been swapped in the cosine.
- Both band rows raise the order: an N th-order low pass becomes a **$2N$ th-order** band pass or band stop.

6.7 Last-minute recall

Say these without thinking

- $\alpha = -20 \log_{10} |H|$, and δ_p means an edge gain of $1 - \delta_p$.
- $N = \frac{\log [(10^{0.1\alpha_s} - 1)/(10^{0.1\alpha_p} - 1)]}{2 \log(\Omega_s/\Omega_p)}$, **round up**.
- $\Omega_c = \Omega_p / (10^{0.1\alpha_p} - 1)^{1/2N}$.
- Bilinear: pre-warp $\Omega = \frac{2}{T} \tan \frac{\omega}{2}$, map back $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$, and **T cancels**.
- Impulse invariance: $\Omega = \omega/T$, and $\frac{1}{s-p_k} \rightarrow \frac{1}{1-e^{p_k T} z^{-1}}$.
- Butterworth poles: radius Ω_c , angles $\pi \frac{N+2k+1}{2N}$, spacing π/N .
- Chebyshev: $\varepsilon = \sqrt{10^{0.1\alpha_p} - 1}$, $\Omega_c = \Omega_p$, poles on an ellipse.
- LP $\rightarrow$ HP: $z^{-1} \rightarrow -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}$, $\alpha = -\frac{\cos \frac{\omega_p + \omega'_p}{2}}{\cos \frac{\omega_p - \omega'_p}{2}}$.
- $\omega = 2\pi f/f_s$ when the spec is in Hz.

Mistakes that cost marks

- **Forgetting to pre-warp** in a bilinear design, or pre-warping in an impulse-invariance one.
- Rounding N down, or to nearest.
- Taking α_s as negative because the paper prints -15 dB. The formula wants $+15$.
- Reading $\delta_p = 0.11$ as a gain of 0.11. It is a **deviation**: the gain is 0.89.
- Using all $2N$ roots of the Butterworth equation. **Only the left half**.
- Substituting $z = e^{sT}$ into $H_a(s)$ for impulse invariance. That is not the method; only the **poles** move that way.
- Giving an even-order Chebyshev filter a DC gain of 1. It is $1/\sqrt{1 + \varepsilon^2}$.
- Claiming impulse invariance can make a high-pass filter. **It cannot**.
- Carrying T through a bilinear design and getting a T -dependent answer. **If T survives, there is an arithmetic error**.

7. Discrete Fourier Transform and FFT

7 hrs · 99 questions across the 45 papers · worth about 14 of 80 marks · 3 syllabus sub-topics · the heaviest chapter in the subject.

Read this first:

- **Three questions, and only three**, in 44 of the 45 papers:
 - **Compute an N -point DFT by DIT or DIF** TOP 43/45, 340 marks. Almost always **8-point**, almost always "draw the butterfly".
 - **Circular convolution** TOP 39/45, 253 marks, in three disguises that are all the same calculation (\$7.5).
 - **Bookwork on the DFT** TOP 17/45, 52 marks: why DFT, why FFT, the properties, zero padding.
- **The FFT is not a new transform**. It is the DFT, computed by a rearrangement that costs $\frac{N}{2} \log_2 N$ multiplications instead of N^2 . Anyone who says "the FFT of $x[n]$ " means its DFT.
- **DIT and DIF give identical answers**. They differ only in where the bit-reversal sits and where the twiddle factor is applied. If a paper does not say which, either is a full-marks answer.
- **Everything rests on two identities**, and nothing else:

$$W_N = e^{-j2\pi/N}, \quad W_N^{k+N/2} = -W_N^k, \quad W_N^{k+N} = W_N^k$$

The first halves the work at every stage; the second is what makes the recursion close.

- **A sequence shorter than N is zero padded**. Eleven papers print six or seven values for an 8-point transform. That is not a misprint to fix, it is the zero-padding they ask you to define elsewhere in the same paper.

7.1 What the DFT is, and why it exists

Why do we need the DFT / FFT although we have the DTFT? Differentiate between the DFT and the DTFT 1+5 2+4 6 2

TOP 13/45 79 Bh, 78 Bh, 71 Ch, 81 Bh, 79 Ch, 70 Bh, 79 Ba, 80 Ba, 74 Ma · asked in nine papers, always the same two reasons

Why the DTFT is not enough

- The DTFT $X(e^{j\omega}) = \sum_n x[n]e^{-j\omega n}$ is a **continuous** function of ω .
- **A computer cannot store a continuous function**, and it cannot sum an infinite series. Neither end of the DTFT is computable.
- The DFT fixes both at once: **finite** sequence in, **finite** set of samples out.

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{nk}, \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-nk}$$

- **The DFT is the DTFT sampled at N equally spaced points**, $\omega_k = 2\pi k/N$. Nothing more.
- The price: the DFT implicitly treats $x[n]$ as **one period of a periodic sequence**. Every oddity in this chapter, circular shift, circular convolution, wrap-around, follows from that one fact.

DFT against DTFT, as a table

	DTFT	DFT
Input	any length, even infinite	finite , N points
Output	continuous in ω	discrete , N points
Periodic in	ω , period 2π	k , period N
Computable	no	yes
Assumes	nothing	$x[n]$ is periodic with period N
Convolution	linear	circular

And why the FFT

- The DFT as defined costs N^2 complex multiplications. **The FFT is an algorithm for the same numbers** at $\frac{N}{2} \log_2 N$.
- For $N = 8$: 64 against 12. For $N = 1024$: 1 048 576 against 5120, a **speed-up of 204.8**. Quote a number; it is worth a mark.

How does the FFT reduce computational complexity compared with direct computation of the DFT?

5 2 4+2

TOP 12/45 the same nine papers, plus 74 Bh 2, 73 Ma 2, 79 Ch 4+2

The argument, in four lines

1. Direct DFT: N outputs, each a sum of N products. N^2 **complex multiplications** and $N(N-1)$ **additions**.
2. Split $x[n]$ into two half-length sequences. Two $\frac{N}{2}$ -point DFTs cost $2(\frac{N}{2})^2 = \frac{N^2}{2}$, **half the work**, plus $\frac{N}{2}$ twiddle multiplications to recombine.
3. **Repeat the split**. $N = 2^m$ allows it $m = \log_2 N$ times, until only 2-point DFTs remain, and a 2-point DFT needs **no multiplication at all** ($W_2^0 = 1$).
4. Each of the $\log_2 N$ stages has $\frac{N}{2}$ butterflies and each butterfly costs one multiplication:

$$\frac{N}{2} \log_2 N \text{ multiplications, } N \log_2 N \text{ additions}$$

The numbers to quote

N	Direct N^2	FFT $\frac{N}{2} \log_2 N$	Speed-up
8	64	12	5.33
32	1024	80	12.8
256	65 536	1024	64
1024	1 048 576	5120	204.8

- **The saving grows with N** , which is the point: it is $2N/\log_2 N$, not a constant factor.
- **Radix-2 needs N a power of 2**. If it is not, zero pad up to one. That is the second reason zero padding exists.
- The two properties being exploited are **periodicity** $W_N^{k+N} = W_N^k$ and **symmetry** $W_N^{k+N/2} = -W_N^k$. Name them: they are usually worth a mark each.

7.2 Properties of the DFT

State and prove the circular convolution property of the DFT; write about the multiplication and convolution properties 2+2+4 6 1

HOT 4/45 70 Ch 2+2+4, 69 Bh 6, 75 Ash 1, 73 Bh 1 · the proof is five lines

The property, and its proof

- **Statement**. If $x_3[n] = x_1[n] \otimes x_2[n]$, the *circular* convolution, then

$$X_3[k] = X_1[k] X_2[k]$$

Circular convolution in time is multiplication in frequency.

- **Proof**. Write the circular convolution and transform it:

$$X_3[k] = \sum_{n=0}^{N-1} \left(\sum_{m=0}^{N-1} x_1[m] x_2[((n-m))_N] \right) W_N^{nk}$$

- Swap the sums and put $\ell = n - m$, so $n = \ell + m$ and $W_N^{nk} = W_N^{mk} W_N^{\ell k}$:

$$= \sum_{m=0}^{N-1} x_1[m] W_N^{mk} \sum_{\ell=0}^{N-1} x_2[((\ell))_N] W_N^{\ell k} = X_1[k] X_2[k]$$

- **The modulo- N index is what makes the inner sum come out as $X_2[k]$ exactly**, with no leftover terms. That is the whole proof, and the step worth pointing at.
- **The dual (multiplication property)**: multiplying in time is *circular convolution divided by N* in frequency, $x_1[n] x_2[n] \longleftrightarrow \frac{1}{N} X_1[k] \otimes X_2[k]$. **Do not forget the $1/N$** .

The full property table (Oppenheim Table 8.2)

Sequence, length N	N -point DFT
$x[n]$	$X[k]$
$ax_1[n] + bx_2[n]$	$aX_1[k] + bX_2[k]$
$X[n]$	$Nx[(-(k))_N]$
$x[((n-m))_N]$	$W_N^{km} X[k]$
$W_N^{-\ell n} x[n]$	$X[((k-\ell))_N]$
$\sum_m x_1[m] x_2[((n-m))_N]$	$X_1[k] X_2[k]$
$x_1[n] x_2[n]$	$\frac{1}{N} \sum_{\ell} X_1[\ell] X_2[((k-\ell))_N]$
$x^*[n]$	$X^*[((-k))_N]$
$x[n]$ real	$X[k] = X^*[((-k))_N]$

- **Row 4 is the circular shift**: a shift in time is a *phase* in frequency and nothing else. The magnitude spectrum does not move.
- **Row 3 is duality**, and it is the whole of the 67 Mng question in §7.3.
- The last row is the free check on every answer in this chapter: **a real sequence has a conjugate-symmetric spectrum**, so $X[N-k] = X^*[k]$. Compute half the values and mirror the rest.

Define zero padding, and Parseval's theorem 1 2

TOP 7/45 74 Ch, 74 Ash, 70 Ch, 81 Ch, 75 Bh, 74 Bh 1; 77 Ch 2 · eight free marks

Zero padding

- **Appending zeros** to a sequence before taking its DFT, to reach a chosen length N .
- **Three reasons**, and papers accept any of them:
 - to reach a **power of 2** so a radix-2 FFT can run;
 - to make $N \geq L + M - 1$ so a **circular convolution equals the linear one** (§7.5, and this is the reason the question sits beside those problems);
 - to **interpolate the spectrum**: more samples of the same underlying DTFT.
- **Zero padding does NOT add information and does NOT improve resolution**. It samples the same DTFT more finely. Saying otherwise loses the mark.

Parseval's theorem

- Energy is the same computed either side of the transform:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$$

- **Note the $1/N$** , which is on the frequency side, matching the IDFT.
- Useful as an arithmetic check: compute $\sum |x[n]|^2$ from the given sequence in two seconds, then confirm the $X[k]$ you just worked out.

Linear against circular convolution (75 Ch asks for it directly)

- **Linear**: indices run freely, result has length $L + M - 1$, and it is what a real LTI system does.
- **Circular**: indices are taken mod N , result has length N , and it is what multiplying two DFTs gives you.
- **They agree exactly when $N \geq L + M - 1$** , and otherwise the circular one is the linear one **wrapped round and added to itself**.

7.3 The IDFT from a forward DFT

Using the duality property, how can a forward DFT routine be used to compute the IDFT? 8 2+6

PIN 2/45 67 Mng 8, 72 Ash 2+6 · 8 marks for four lines of algebra

The conjugate trick, which is the expected answer

1. Start from the IDFT and conjugate both sides:

$$x[n] = \frac{1}{N} \sum_k X[k] W_N^{-nk} \Rightarrow x^*[n] = \frac{1}{N} \sum_k X^*[k] W_N^{nk}$$

2. **The right-hand sum is a forward DFT of $X^*[k]$** . So:

$$x[n] = \frac{1}{N} \left(\text{DFT} \{ X^*[k] \} \right)^*$$

3. **Three steps**: conjugate the input, run the same FFT, conjugate the output and divide by N .
4. **No second routine is needed**, which is the point of the question: the same butterfly hardware computes both directions.

The duality route, if the paper insists

- Duality says $X[n] \xrightarrow{\text{DFT}} Nx[((-k))_N]$.
- So feeding the *spectrum* $X[k]$ into a forward DFT returns the original sequence, scaled by N and **index reversed modulo N** :

$$x[n] = \frac{1}{N} \text{DFT} \{ X[k] \} \Big|_{k \rightarrow -n \bmod N}$$

- Index reversal modulo N leaves entry 0 alone and reverses the rest, so $\{a, b, c, d\} \rightarrow \{a, d, c, b\}$. **That single sentence is what the 8 marks are for.**
- Both routes are correct. The conjugate one is easier to defend and is what 72 Ash's worked answer needs (§2 of the numericals).

7.4 The FFT: decimation in time and in frequency

How do you use the DIT-FFT algorithm to compute the 8-point DFT of a sequence? Explain with the flow graph **6 4**

TOP 43/45 70 Ma **6**, 69 Bh **4** (4-point), and the 41 computation papers in \$7.6

Decimation in time, derived

- Split the sum by **even and odd** n :

$$X[k] = \sum_r x[2r]W_N^{2rk} + \sum_r x[2r+1]W_N^{(2r+1)k}$$

- $W_N^2 = W_{N/2}$, so both sums are $\frac{N}{2}$ -point DFTs:

$$X[k] = G[k] + W_N^k H[k]$$

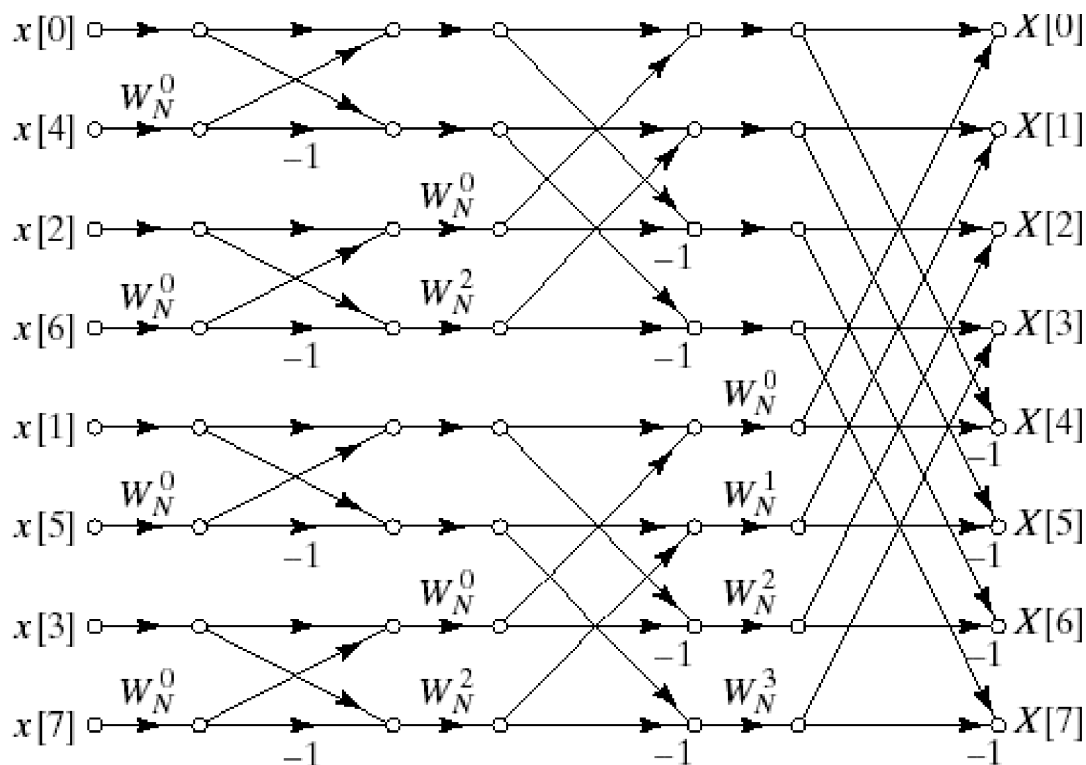
with G from the even samples and H from the odd ones.

- The other half comes free from the symmetry:

$$X[k + \frac{N}{2}] = G[k] - W_N^k H[k]$$

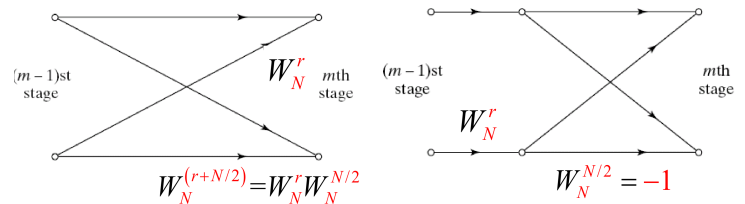
- One multiplication serves two outputs.** That pair is the butterfly, and it is the entire saving.
- Recurse $\log_2 N$ times. Splitting by even and odd repeatedly leaves the input in **bit-reversed order**; the output comes out natural.

The 8-point DIT flow graph (**draw this one**)



Input bit-reversed $\{x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]\}$, output natural. Three stages. Twiddles W_N^0 in stage 1; W_N^0, W_N^2 in stage 2; $W_N^0, W_N^1, W_N^2, W_N^3$ in stage 3. The twiddle count doubles each stage and the butterfly span doubles with it.

The butterfly



Left, the butterfly as first drawn: two twiddles. Right, the same butterfly after $W_N^{r+N/2} = W_N^r W_N^{N/2} = -W_N^r$ is used, so one multiplication and a sign.

Always draw the right-hand form.

- Every stage is $\frac{N}{2}$ of these; there are $\log_2 N$ stages.
- The -1 on the lower branch is not decoration.** Omitting it is the single most common way to lose the marks on a flow graph.

Draw the flow graph for the 8-point decimation in frequency algorithm 8 2+6

TOP 20/45 the DIF half of §7.6: 80 Bh, 76 Ch, 76 Ash, 79 Ba, 73 Ch, 69 Ch, 74 Ash, 70 Bh, 81 Ch, 78 Ch, 77 Ch, 75 Bh, 72 Ma, **68 Bh**, 82 Ba, 80 Ba, 70 Ch, 74 Ch

Decimation in frequency, derived

- This time split the **input** into first and second halves, and the **output** into even and odd k :

$$g[n] = x[n] + x[n + \frac{N}{2}], \quad h[n] = (x[n] - x[n + \frac{N}{2}])W_N^n$$

$$X[2r] = \text{DFT}_{N/2}\{g[n]\}, \quad X[2r+1] = \text{DFT}_{N/2}\{h[n]\}$$

- **The twiddle multiplies the DIFFERENCE, after the subtraction.** In DIT it multiplies before the add. That is the only real difference between the two.
- Recurse $\log_2 N$ times. Input stays natural; **output comes out bit-reversed** and has to be unscrambled.

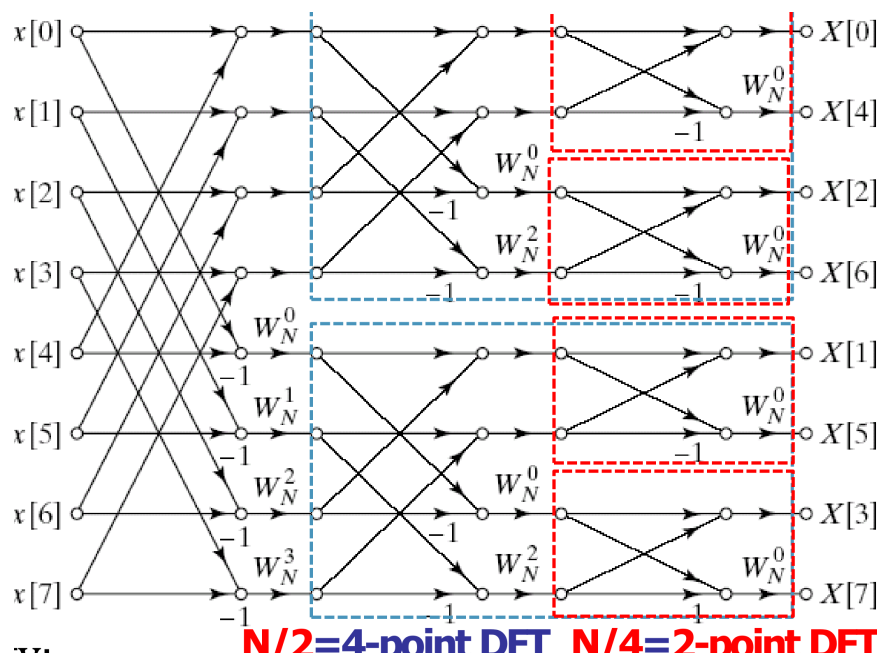
Bit reversal, which is all either algorithm scrambles

- **The same permutation serves both:** 0, 4, 2, 6, 1, 5, 3, 7 for $N = 8$. It is its own inverse, so there is only one order to memorise.
- Get it by writing the index in binary and reading it **backwards**: $3 = 011 \rightarrow 110 = 6$, $1 = 001 \rightarrow 100 = 4$.
- **Exactly one end is scrambled**, never both and never neither. DIT scrambles the input, DIF the output.
- For $N = 4$ it is 0, 2, 1, 3; for $N = 16$, 0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15.
- The scrambling is not overhead you can avoid. It is what lets every butterfly write its result back over its own inputs, so the whole transform runs in **one array with no scratch space**.

DIT against DIF, the whole comparison

	DIT	DIF
Splits	input, even/odd n	output, even/odd k
Input order	bit-reversed	natural
Output order	natural	bit-reversed
Twiddle	before the add	after the subtract
Stages	$\log_2 N$	$\log_2 N$
Multiplies	$\frac{N}{2} \log_2 N$	$\frac{N}{2} \log_2 N$
Answer	identical	

The 8-point DIF flow graph (the other one to be able to draw)



Input natural, **output bit-reversed**: $X[0], X[4], X[2], X[6], X[1], X[5], X[3], X[7]$ read down the right edge. The dashed boxes show the recursion, the $N/2 = 4$ -point DFTs inside the whole transform and the $N/4 = 2$ -point DFTs inside those. The twiddles run W_N^0 to W_N^3 in stage 1 and halve their range at each stage, the mirror image of DIT, where they start at W_N^0 alone and double.

7.5 Circular convolution, and the three ways it is asked

Compute the circular convolution of two sequences 5 6 7 2+6

TOP 39/45 15 papers ask it directly, 12 ask for $x_3 = \text{IDFT}(X_1 X_2)$, 4 ask for a linear convolution done circularly. All 31 are one calculation.

In one line: **circular convolution is ordinary convolution with the index taken modulo N** , so the tail that would have run off the end wraps round and adds onto the front instead. Everything below is that sentence applied.

The three disguises

1. "Compute the circular convolution of x_1 and x_2 ." Direct.
2. "Find $x_3[n]$ if $X_3[k] = X_1[k]X_2[k]$." By the property in §7.2 this **IS the circular convolution**, so do the same arithmetic and do not compute a single DFT.
3. "Find the linear convolution using circular convolution." Zero pad both to $N \geq L + M - 1$, then the same arithmetic again.

Recognising that (2) needs no transform is worth several minutes in the exam. The long route, three DFTs and an IDFT, earns the same marks and takes four times as long. Say in one line that you are using the property, then convolve.

Choosing N

- If the paper names N , use it, and pad both sequences up to it.
- If not, $N = \max(L, M)$ for a circular answer, and $N = L + M - 1$ for a linear one.
- **N smaller than $L + M - 1$ wraps**, and the wrap is not an error: it is the correct circular answer to the question that was asked.
- **N larger than $L + M - 1$ only appends zeros** to the result. Harmless, and sometimes needed to reach a power of 2.

The matrix method, which is the fastest by hand

- Write the **circulant** of one sequence: first column is x_2 , and each next column is the previous rotated *down* by one.

$$y[n] = \sum_{m=0}^{N-1} x_1[m] x_2[(n-m)_N]$$

- Multiply by x_1 as a column vector. **Each row of the product is one output sample**, so the whole answer falls out of one $N \times N$ table.
- Check: $\sum y[n] = (\sum x_1)(\sum x_2)$, always. It is $Y[0] = X_1[0]X_2[0]$ in disguise, it costs two additions, and it catches almost every slip.

The concentric-circle picture

- The same arithmetic, drawn: x_1 clockwise on an inner circle, x_2 anticlockwise on an outer one, multiply the aligned pairs and sum, then rotate one circle a step for the next output.
- **Use the matrix instead.** The circles are slower, easier to misalign, and no paper in the 45 demands them.

7.6 Last-minute recall

Say these without thinking

- $X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}$, $W_N = e^{-j2\pi/N}$, IDFT carries $1/N$ and W_N^{-nk} .
- W_4 : 1, $-j$, -1 , $+j$. $W_8^1 = \frac{1-j}{\sqrt{2}}$, $W_8^2 = -j$, $W_8^3 = \frac{-1-j}{\sqrt{2}}$.
- $W_N^{k+N/2} = -W_N^k$ and $W_N^{k+N} = W_N^k$.
- Bit reversal for $N = 8$: 0, 4, 2, 6, 1, 5, 3, 7.
- FFT cost $\frac{N}{2} \log_2 N$ multiplies, $N \log_2 N$ adds; $N = 1024$ gives $204.8 \times$.
- DIT: bit-reversed in, natural out, twiddle before the add. DIF: natural in, bit-reversed out, twiddle after the subtract.
- $X_3[k] = X_1[k]X_2[k] \Leftrightarrow$ **circular** convolution.
- Linear = circular when $N \geq L + M - 1$.
- $X[0] = \sum x[n]$, and for real x , $X[N-k] = X^*[k]$.
- Parseval: $\sum |x[n]|^2 = \frac{1}{N} \sum |X[k]|^2$.

Mistakes that cost marks

- **Forgetting the -1** on the lower butterfly branch.
- Feeding DIT in natural order, or reading DIF's output in natural order. **Exactly one of the two ends is scrambled, never both and never neither.**
- Using $e^{+j2\pi/N}$ in the forward transform. **The forward sign is minus.**
- Computing three DFTs and an IDFT for a question that only wanted a circular convolution.
- Treating a 6 or 7 entry sequence as an error. **Pad it and say you padded it.**
- Claiming zero padding improves frequency resolution. It does not.
- Dropping the $1/N$ in the IDFT, or in the multiplication property.
- Giving a circular convolution of length $L + M - 1$. **It has length N .**
- Forgetting that $X[k]$ repeats: $X[5]$ of a 4-point DFT is $X[1]$, which is exactly what 75 Ch is testing.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. → reads “leads to / therefore”, + reads “and”, = reads “is”. ✓ means the property holds, × that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months**: 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II). Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 **Bh** is EX 753 Regular, 74 **Ma** is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.